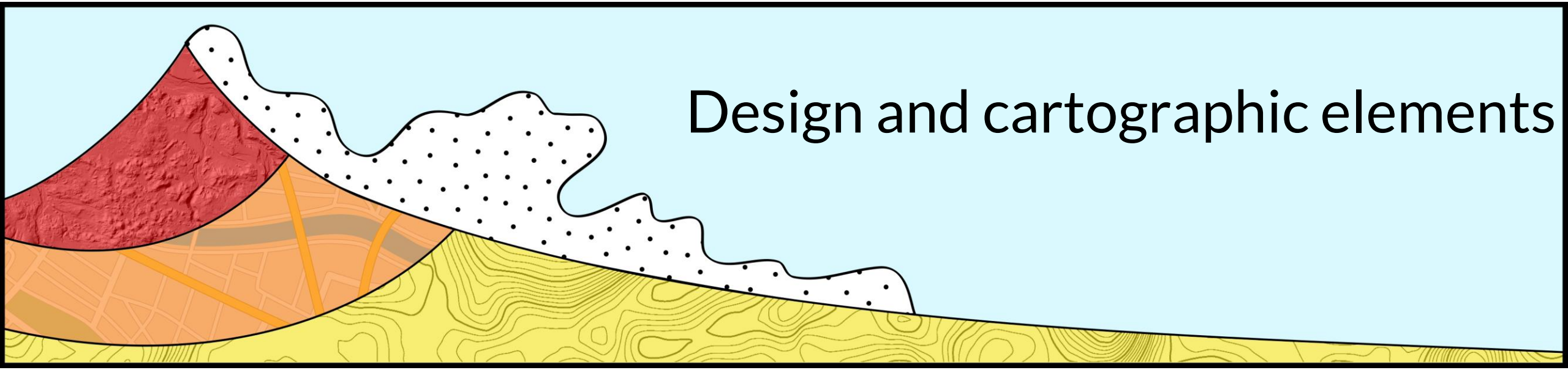
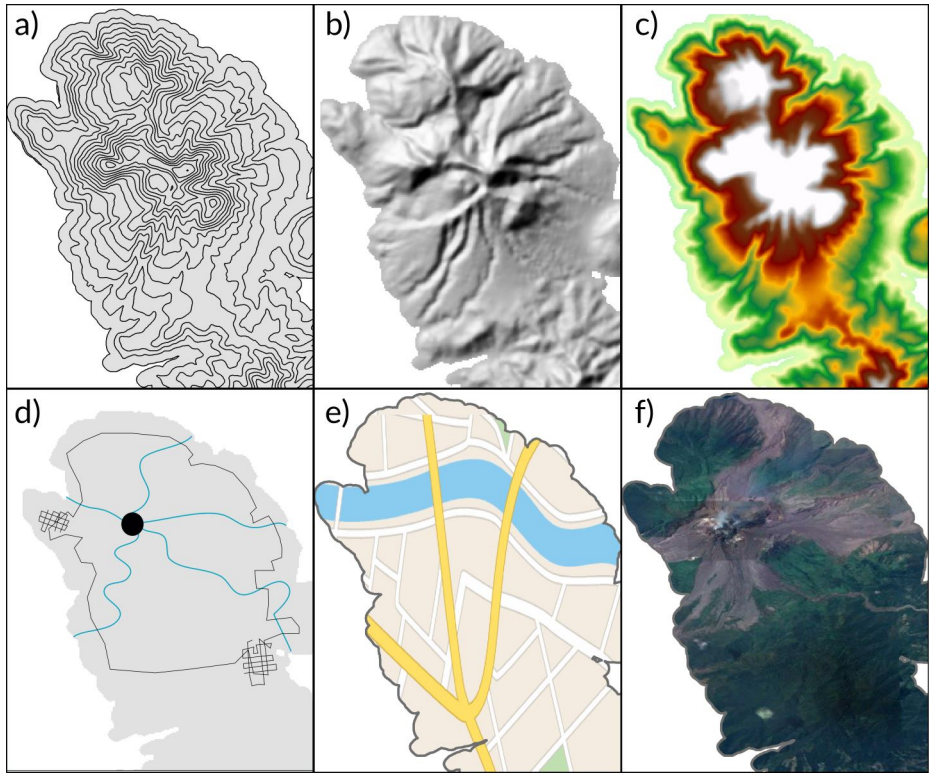
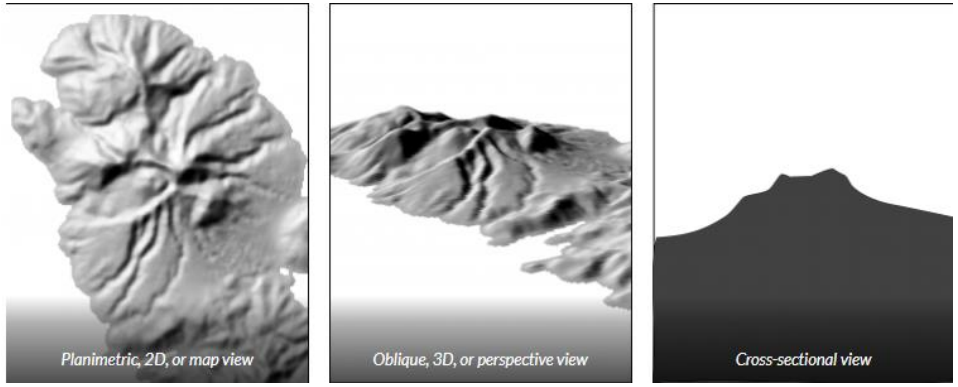
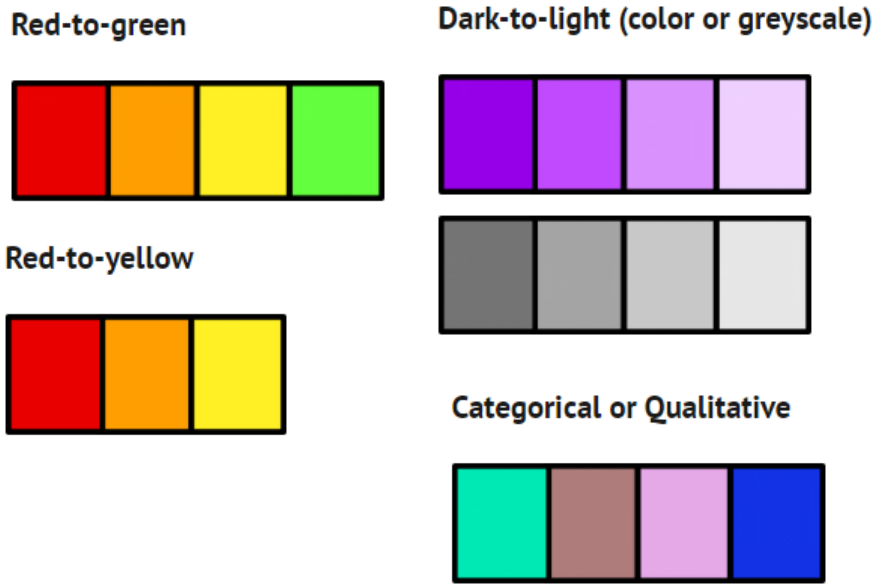


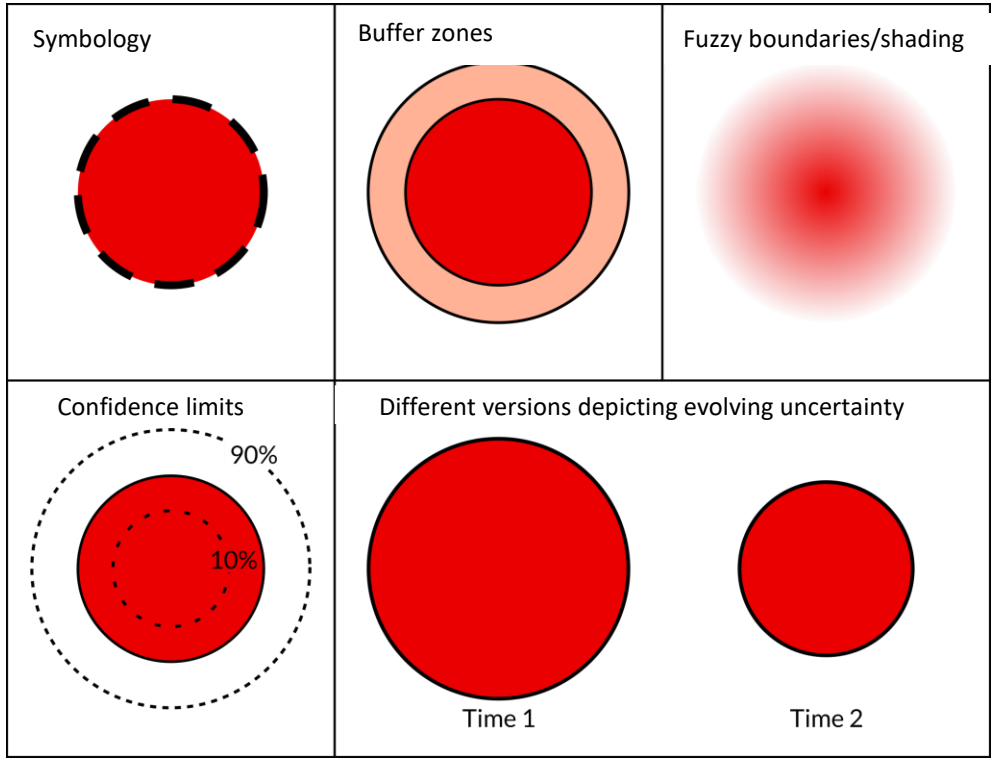
Design and cartographic elements

Design elements









Color and contrast

- The use of color helps to delineate shapes on maps, because human eyes are less sensitive to shapes than to color.

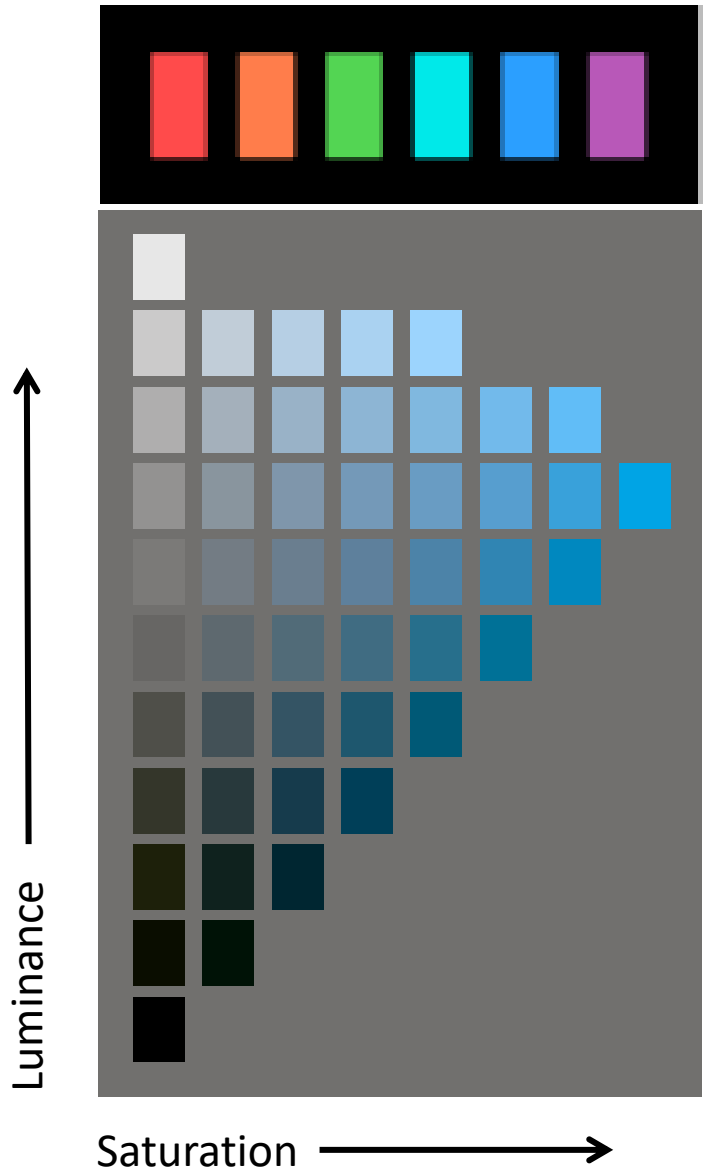
- For this reason, the use of contrasting colors can help users identify the edges of shapes [Keller and Keller 1993].

- The good use of color can positively influence the ability of a map to communicate hazard information, while poor use creates confusion.

http://colorbrewer2.org/
colorusage.arc.nasa.gov NASA AMES Color Usage Research Lab
https://eagereyes.org/basics/rainbow-color-map

Color basics

- Hue: what most people mean by ‘color’
- Saturation: ‘paleness’ or ‘vividness’, distance from grayscale
- Brightness/Luminance: ‘tone’, low values are ‘dark’, high values are ‘light’



Red-to-green



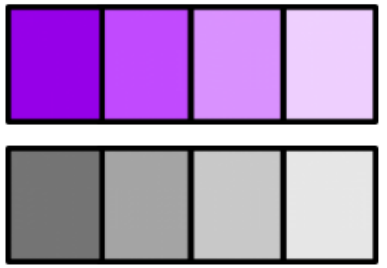
- The logical ordering of sequential red-to-green “stoplight” color schemes (red means danger, yellow means caution, and green means safe) is fairly universal among cultures
- This makes red-to-green color schemes easily understood for relative hazard levels
- Red-to-green color scales may incorrectly imply that green zones are ‘safe’ rather than just the lowest hazard
- Red-to-green color schemes can pose issues for individuals with red-green colorblindness

Red-to-yellow



- Most commonly used
- Users are able to quickly order hazard zones from highest to lowest hazard
- No zones are misinterpreted to be ‘safe zones’ (as is the case when using the color green)
- More easily understood by people with color-blindness

Dark-to-light (color or greyscale)



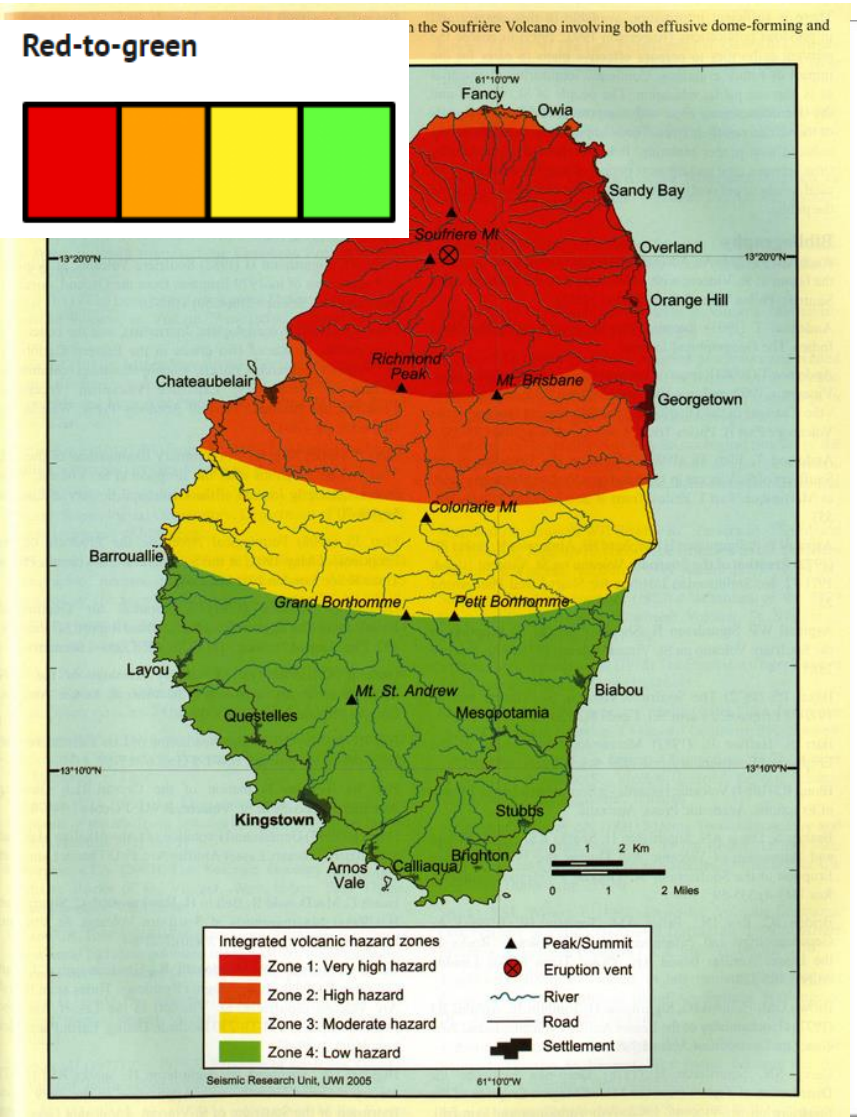
- Dark-to-light sequential color schemes are also well-suited for conveying the relative degree of hazard for different hazard zones
- Most effective when the darkest or most saturated hue represents high values
- Appropriate for the color-blind

Categorical or Qualitative

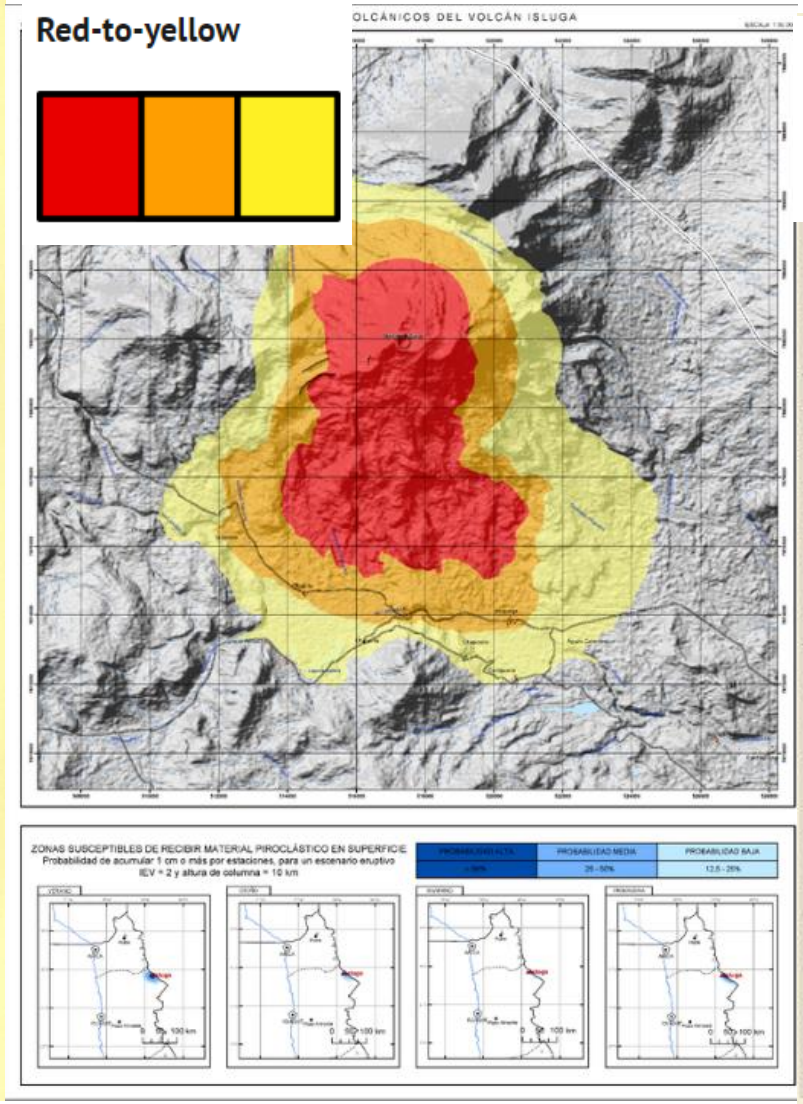


- Sequential hazard levels (high-med-low) are more difficult to discern on maps using categorical or qualitative color schemes, as the map reader must closely study the map legend
- However, well-selected categorical colors may be appropriate for maps displaying non-ordered or categorical data, as is the case for hazard process-focused maps

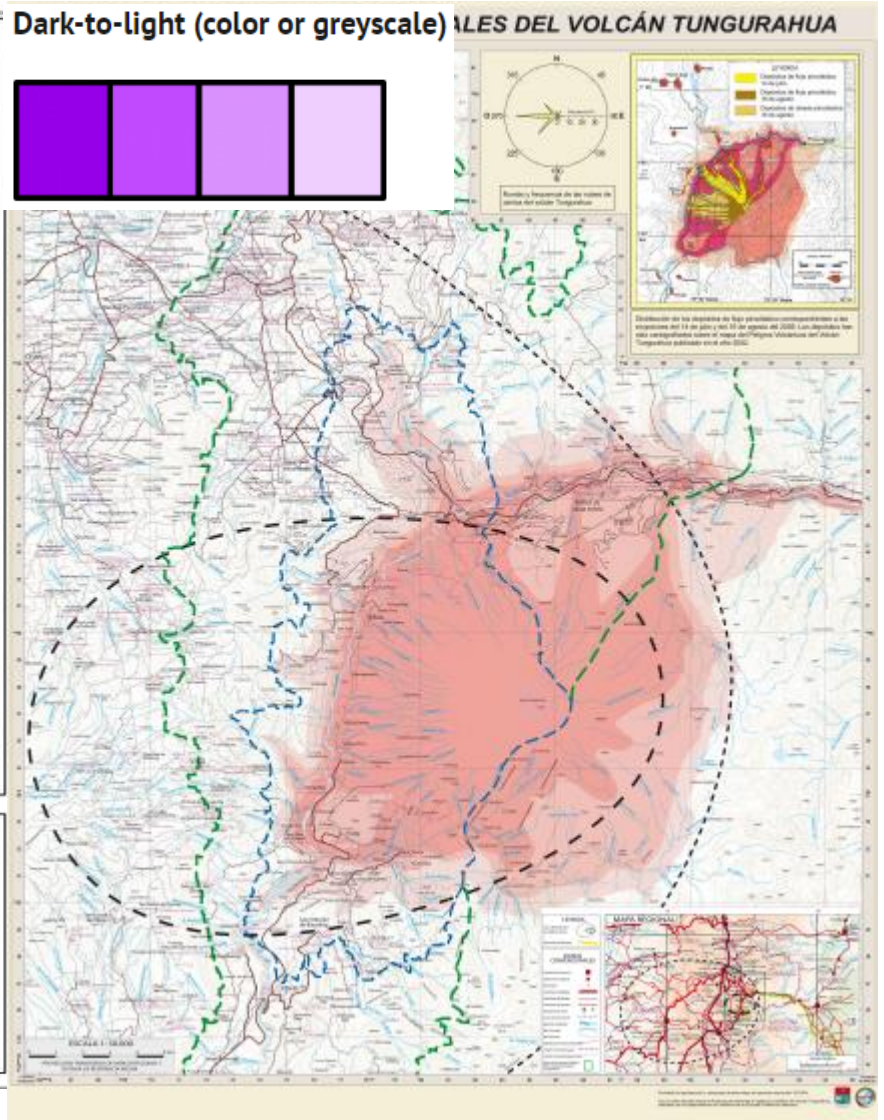
Hazard level-focused (integrated)



Soufrière St. Vincent (Robertson, 2005)

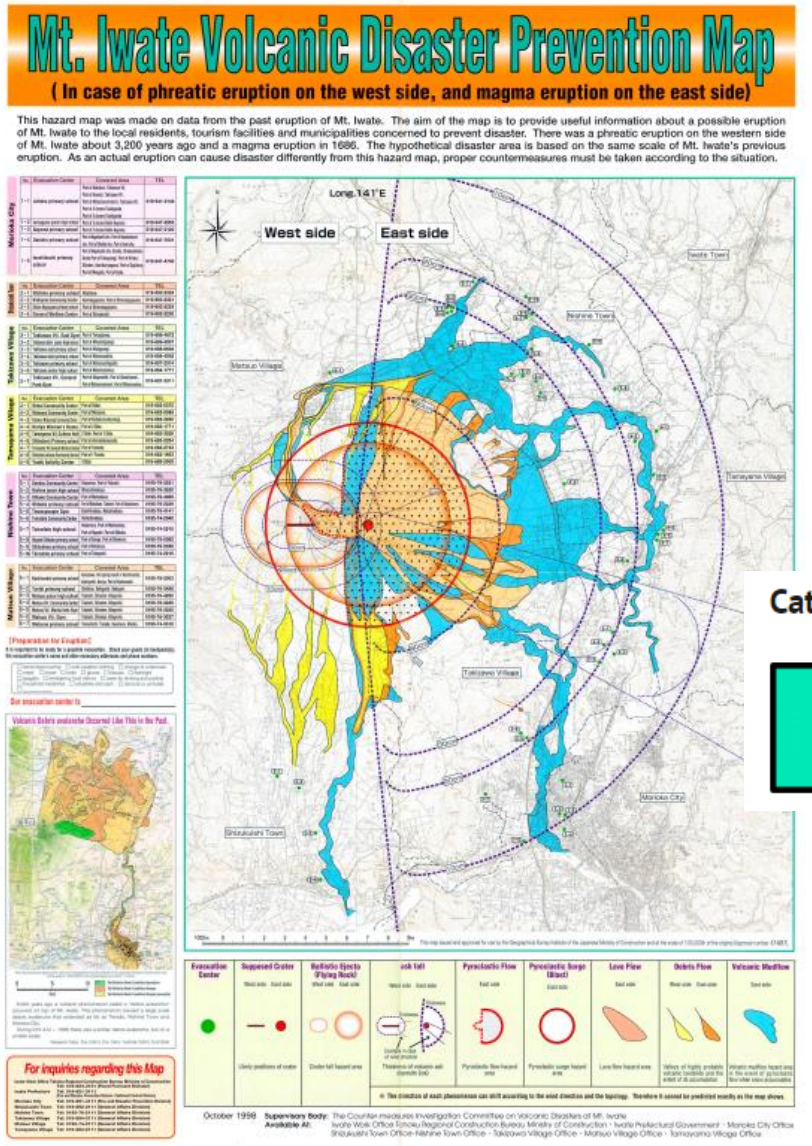


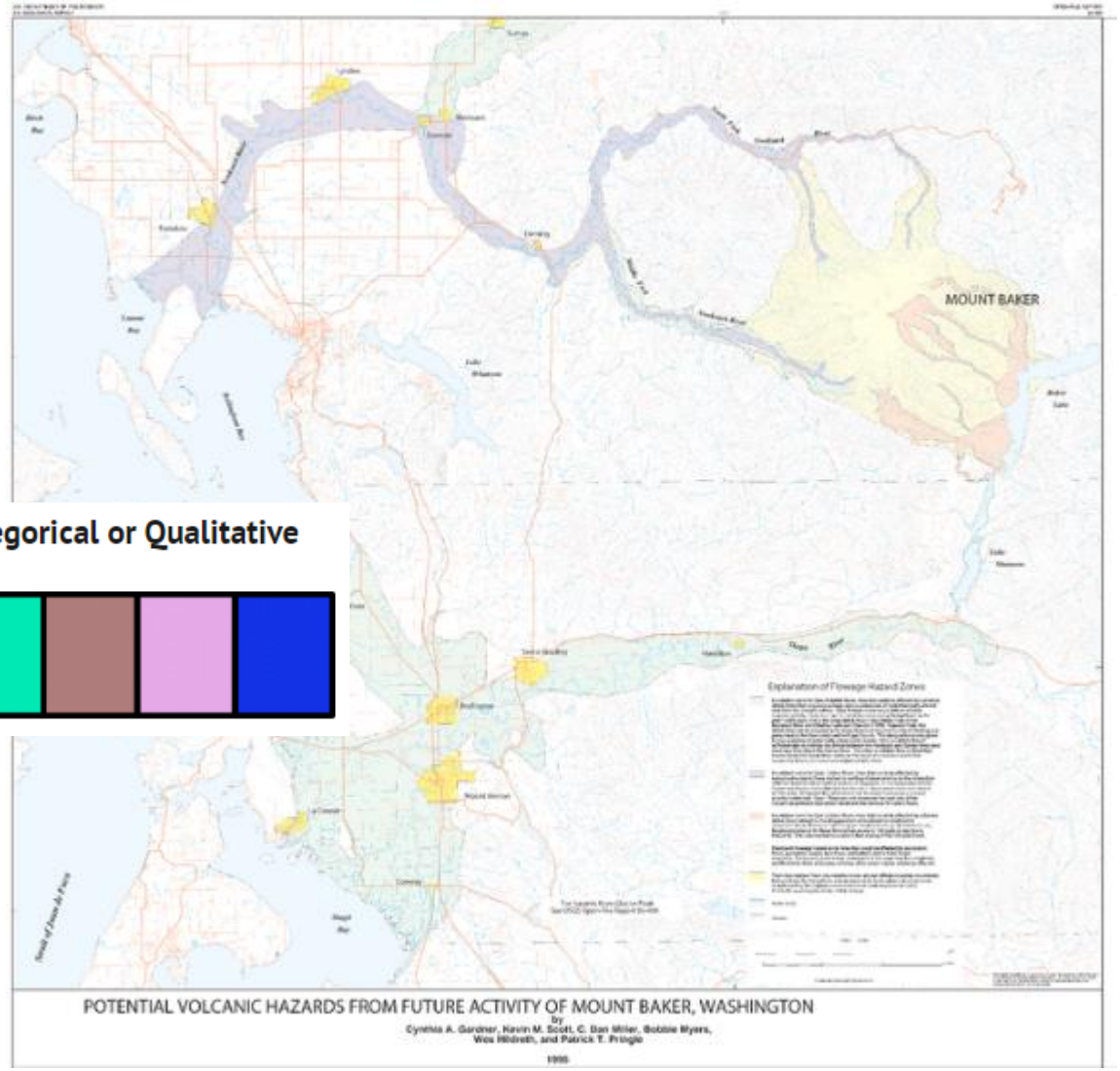
Isluga, Chile (Bertin & Amigo, 2013)



Tungurahua, Ecuador (Samaniengo et al., 2008)

Hazard process-focused (separated)





Categorical or Qualitative



Iwatesan, Japan (1998)

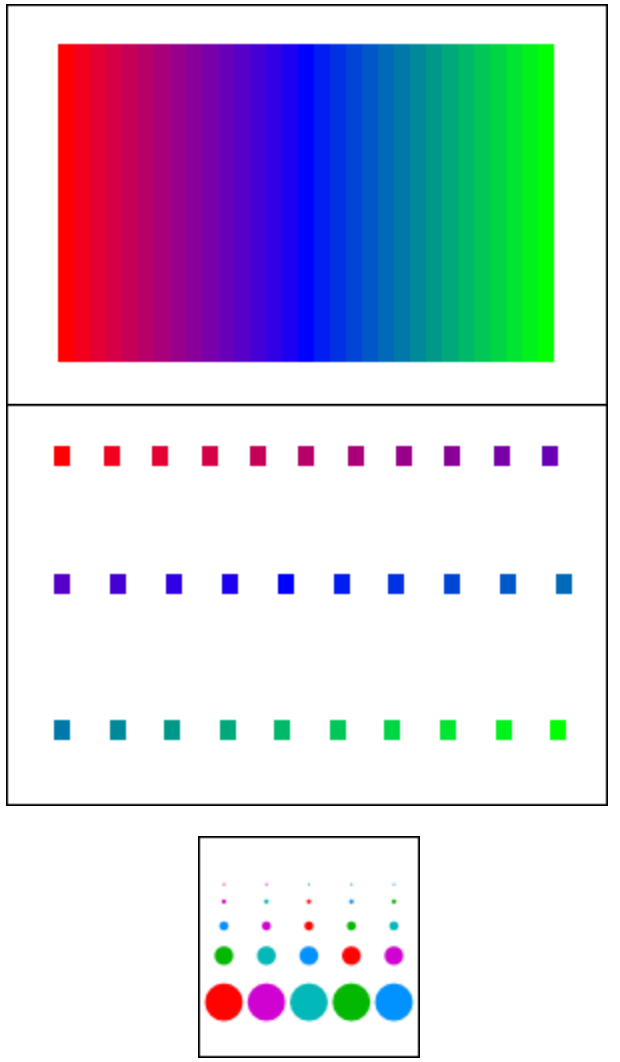
Baker, United States (Gardner et al., 1995)

Color-coding

Small color differences can be distinguished when the areas to be discriminated are large, immediately adjacent to each other and are displayed at the same time.

These conditions occur, for example, in visualization of quantized continuous data as in maps of weather, temperatures, pressures or terrain.

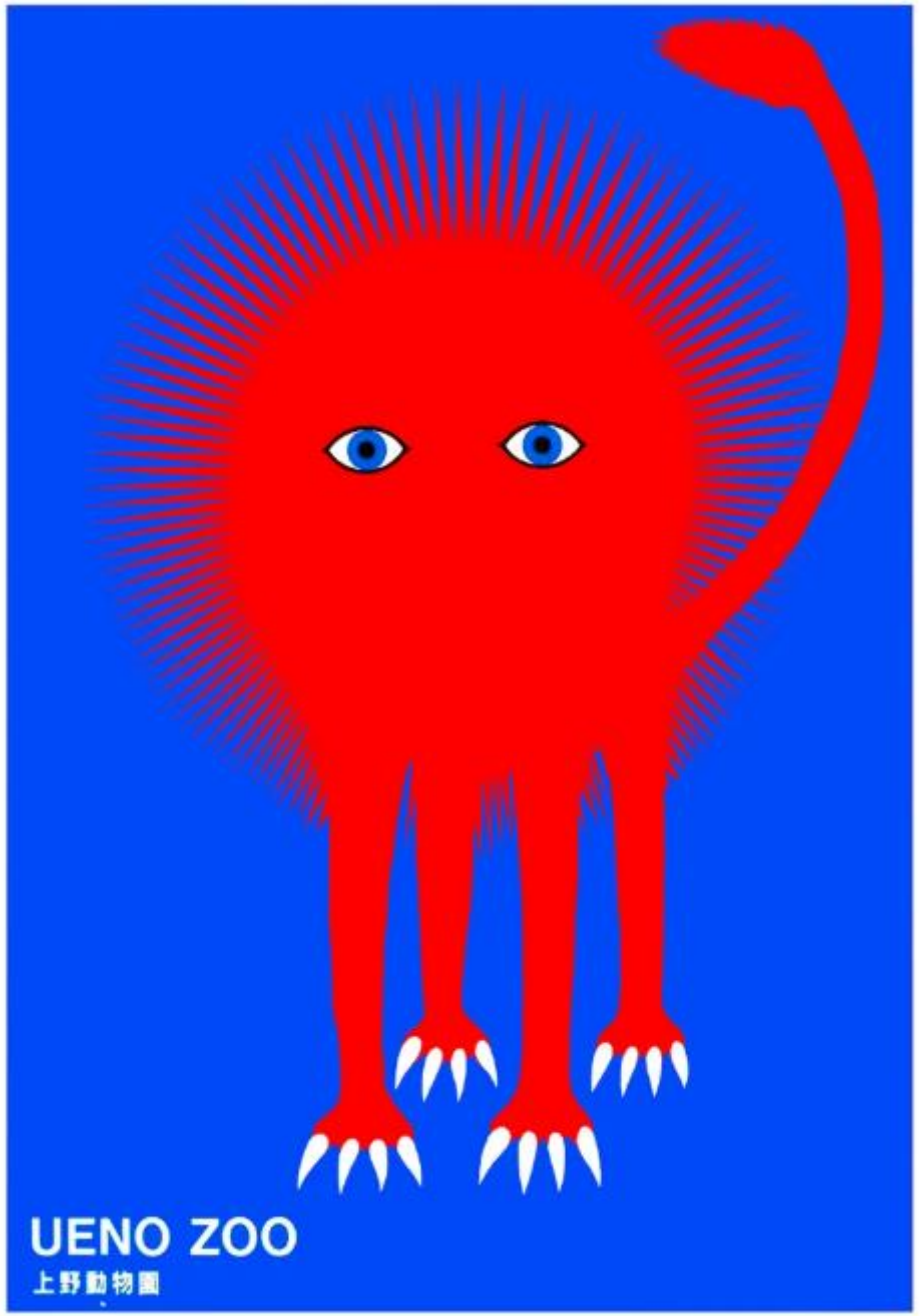
Larger color differences are required whenever conditions deviate from this optimum.



Isoluminance



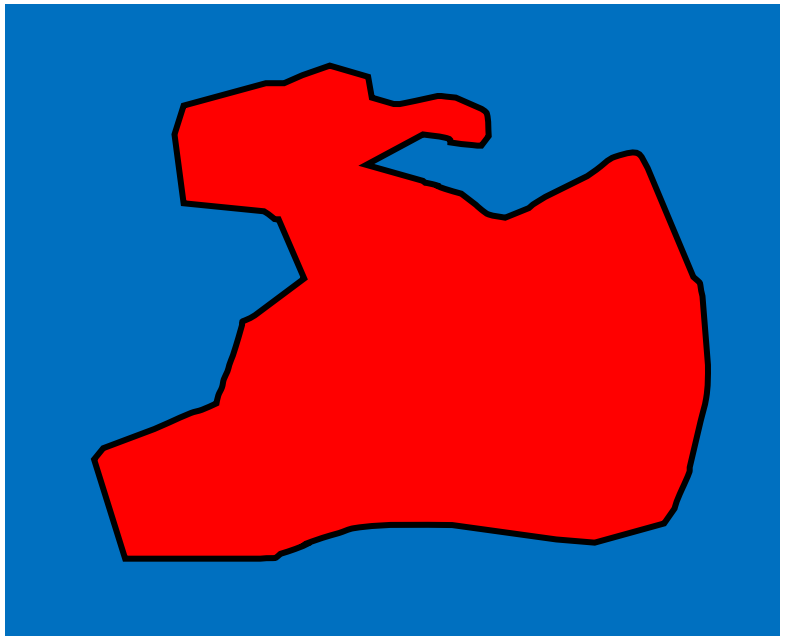
Claude Monet


UENO ZOO
上野動物園

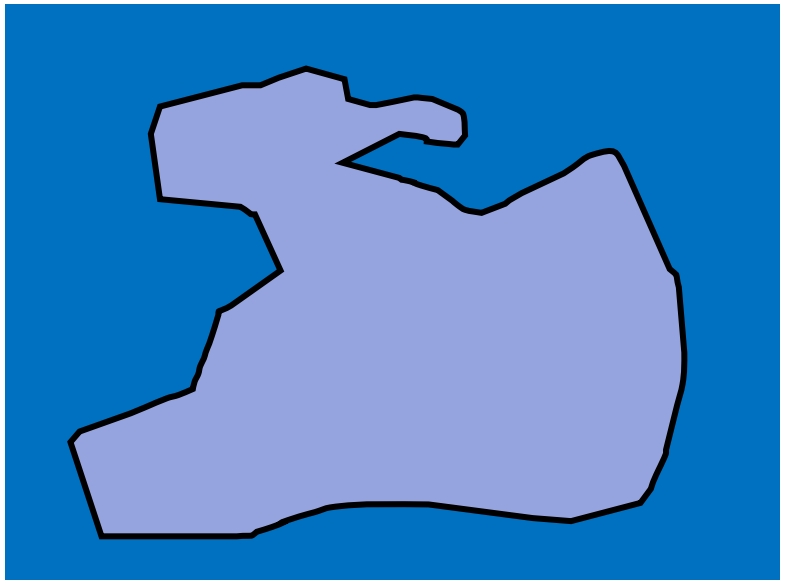


Isoluminance

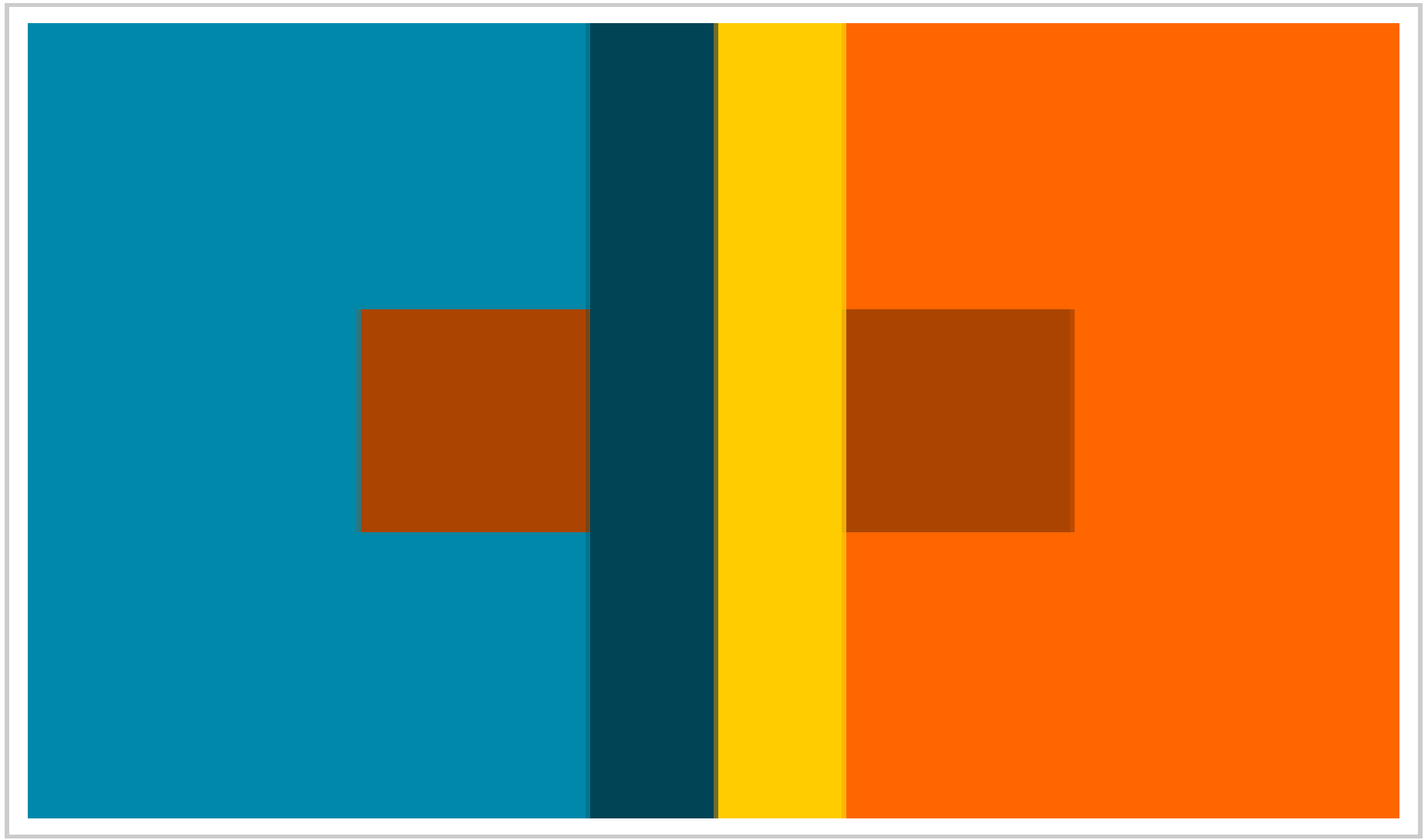








Simultaneous contrast



Simultaneous contrast





Fig. 1. The rainbow color map. Know thy enemy.

Image from K. Moreland

tiny.cc/endrainbow

#endrainbow

http://presentations.copernicus.org/EGU2016-3028_presentation.pdf

Slide courtesy Ed Hawkins, EGU 2016 Making Better Scientific Graphics



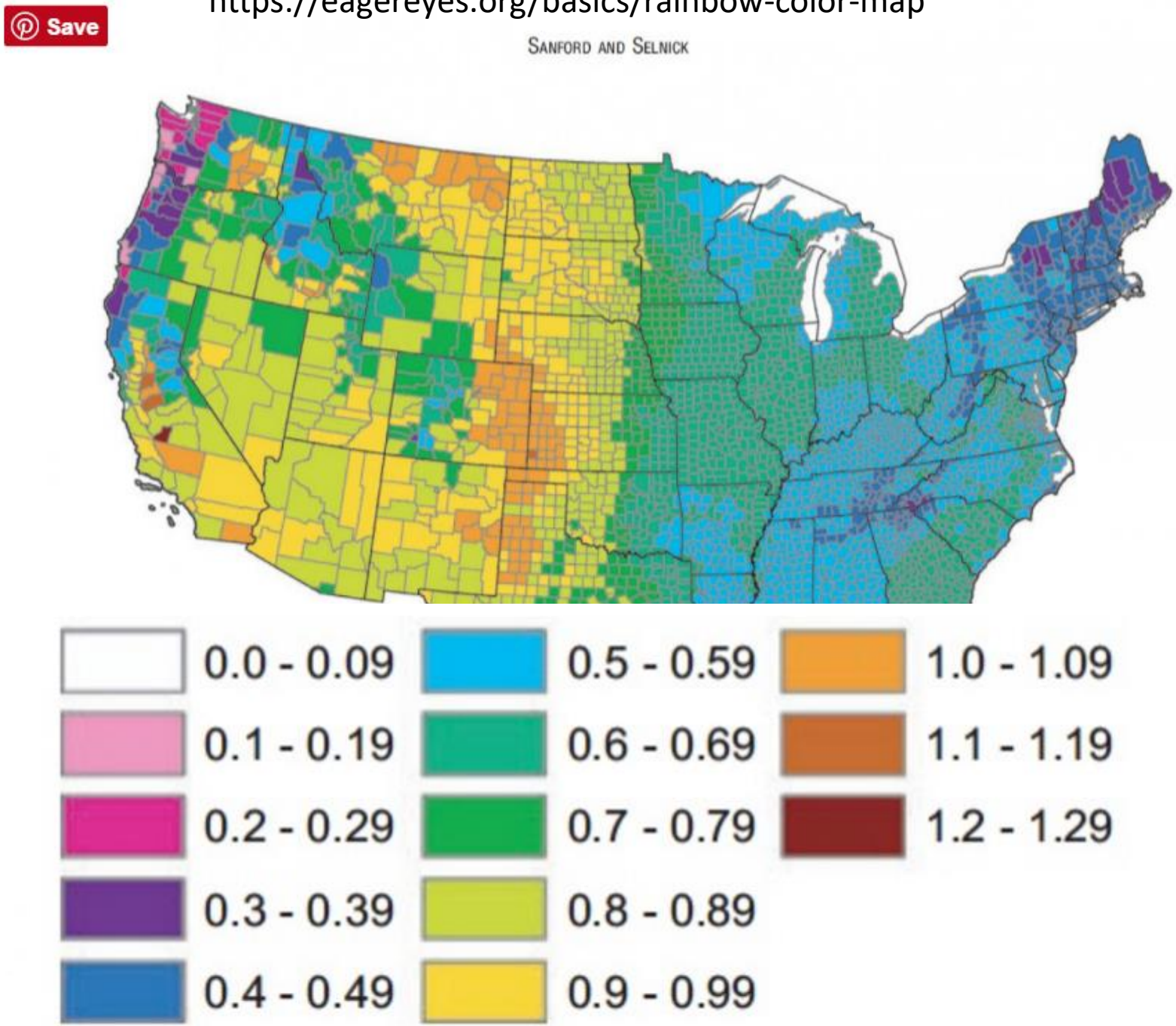
The end of the rainbow

November 18, 2014 communication, journals, visualisation Ed Hawkins

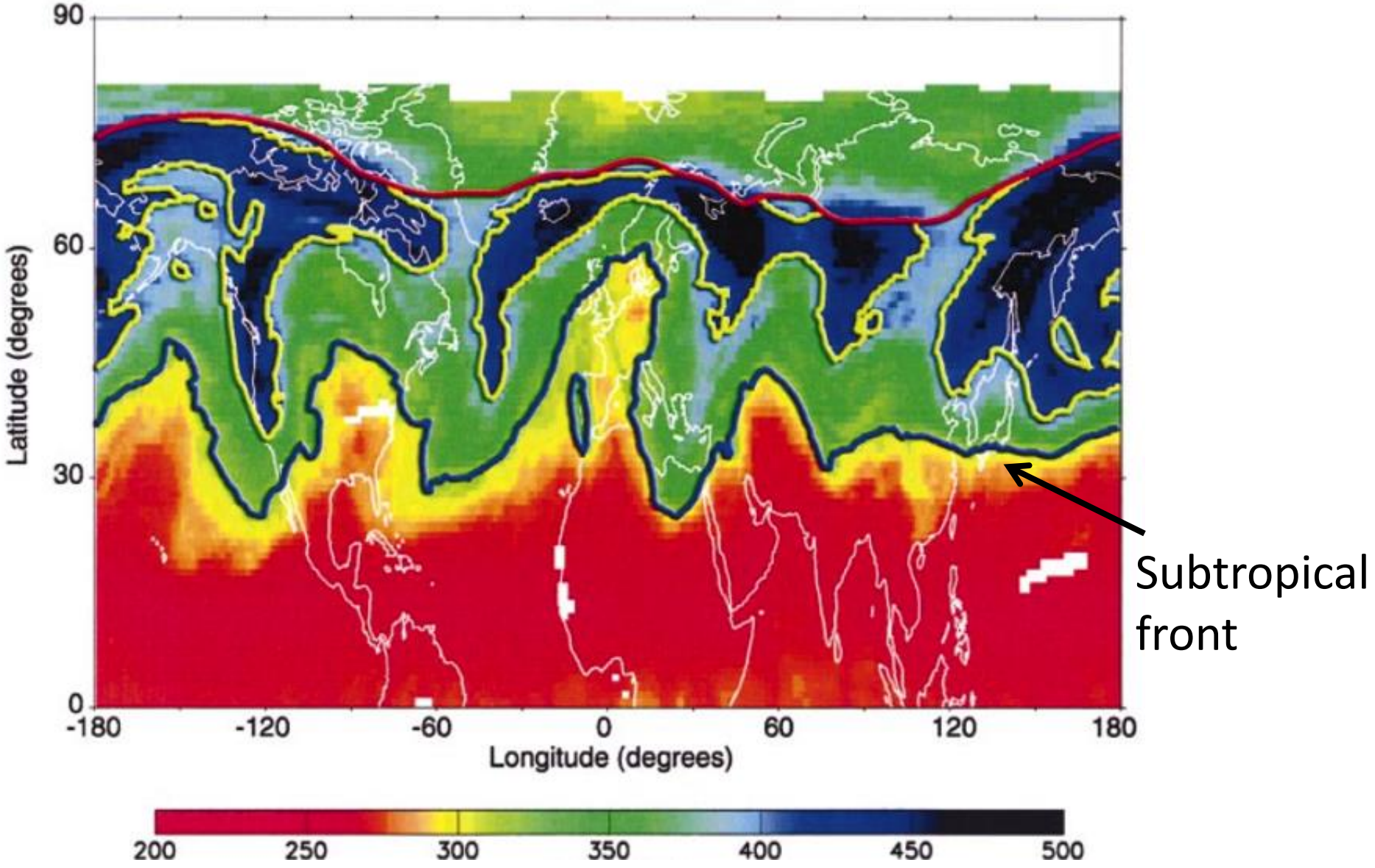
An open letter to the climate science community

Ed Hawkins, Doug McNeall, David Stephenson, Jonny Williams & Dave Carlson

https://eagereyes.org/basics/rainbow-color-map



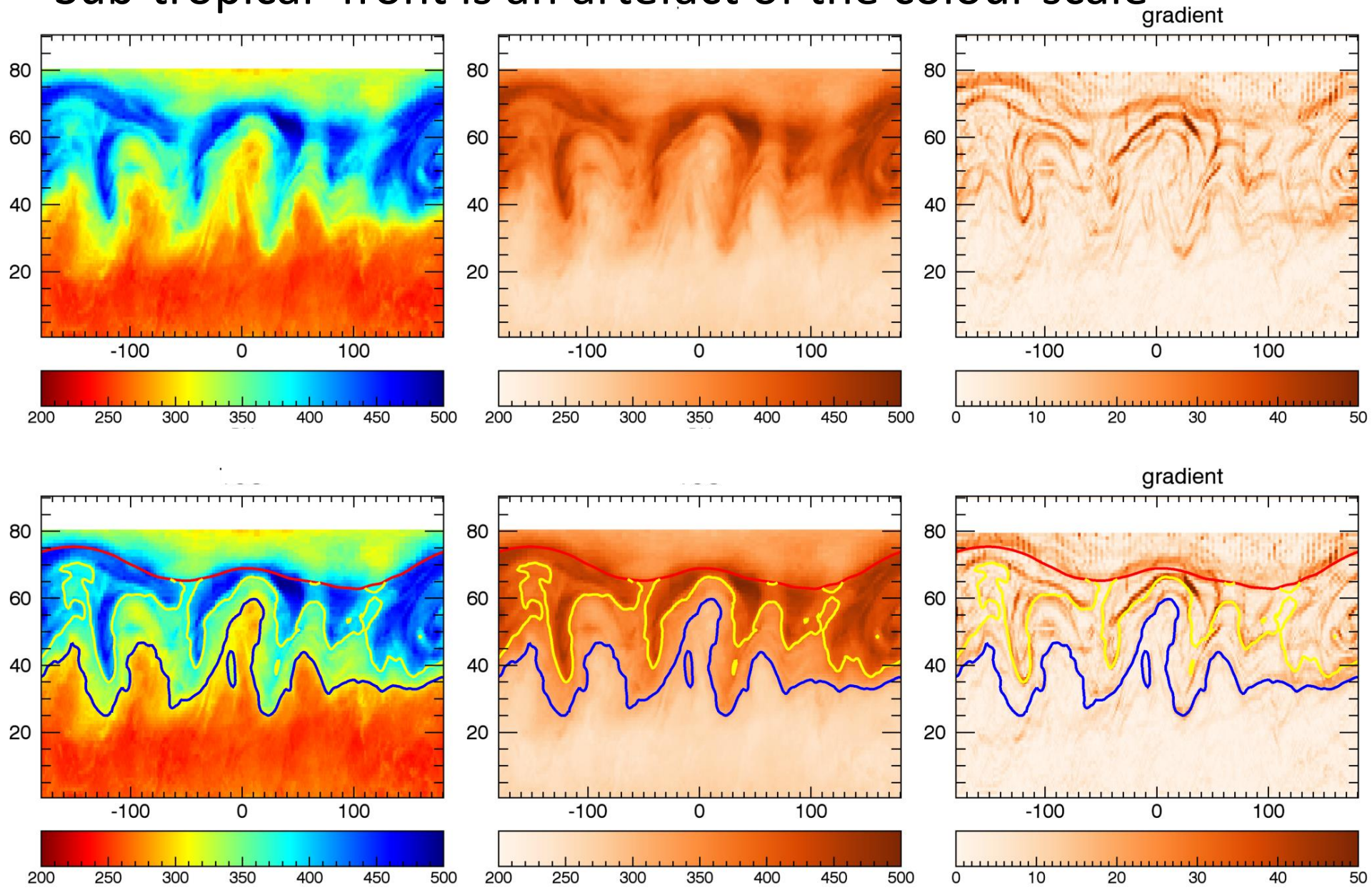
Dangers of rainbow colour scales



Subtropical front

Slide courtesy Ed Hawkins, EGU 2016 Making Better Scientific Graphics

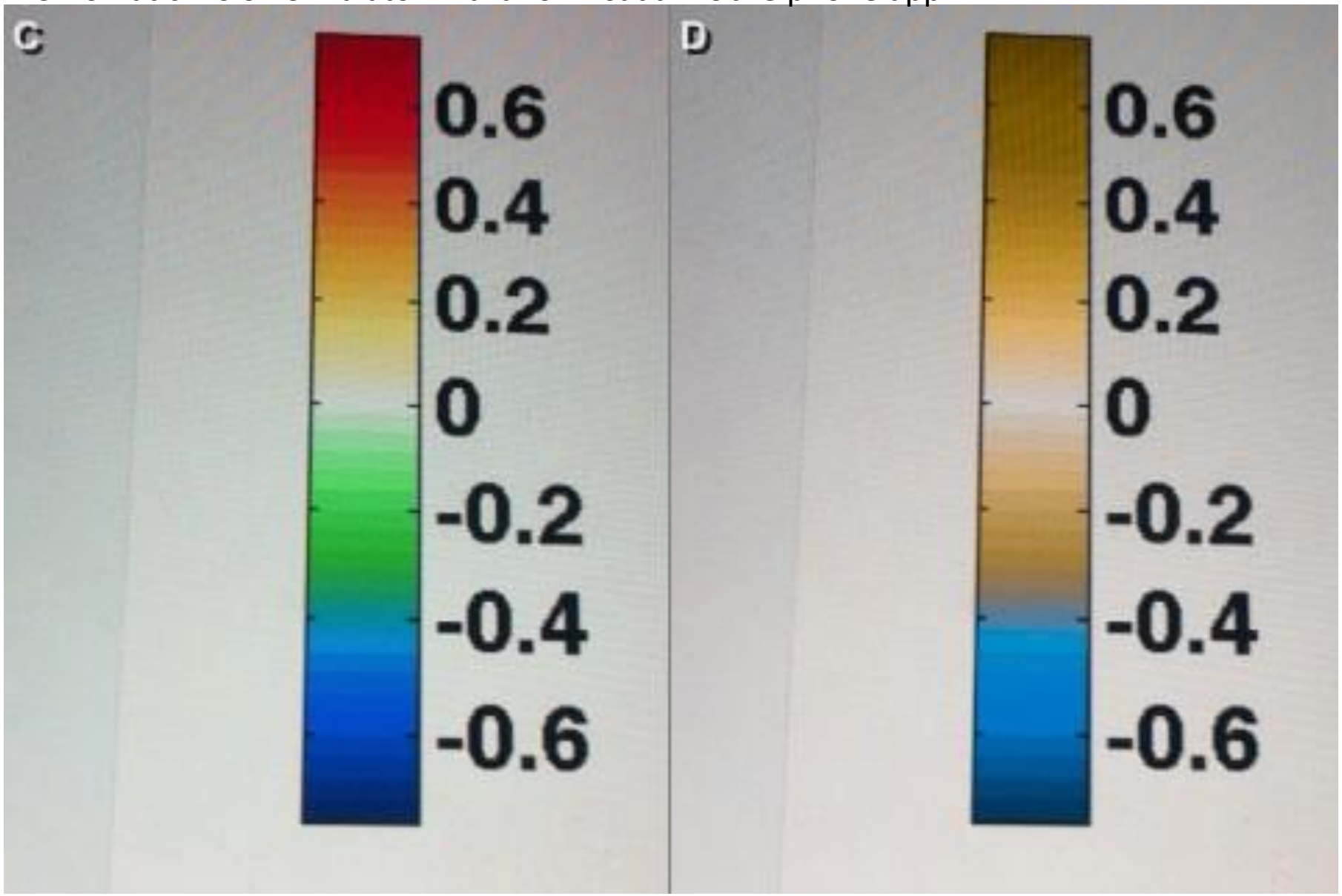
‘Sub-tropical’ front is an artefact of the colour scale



Figures from Sean Davis

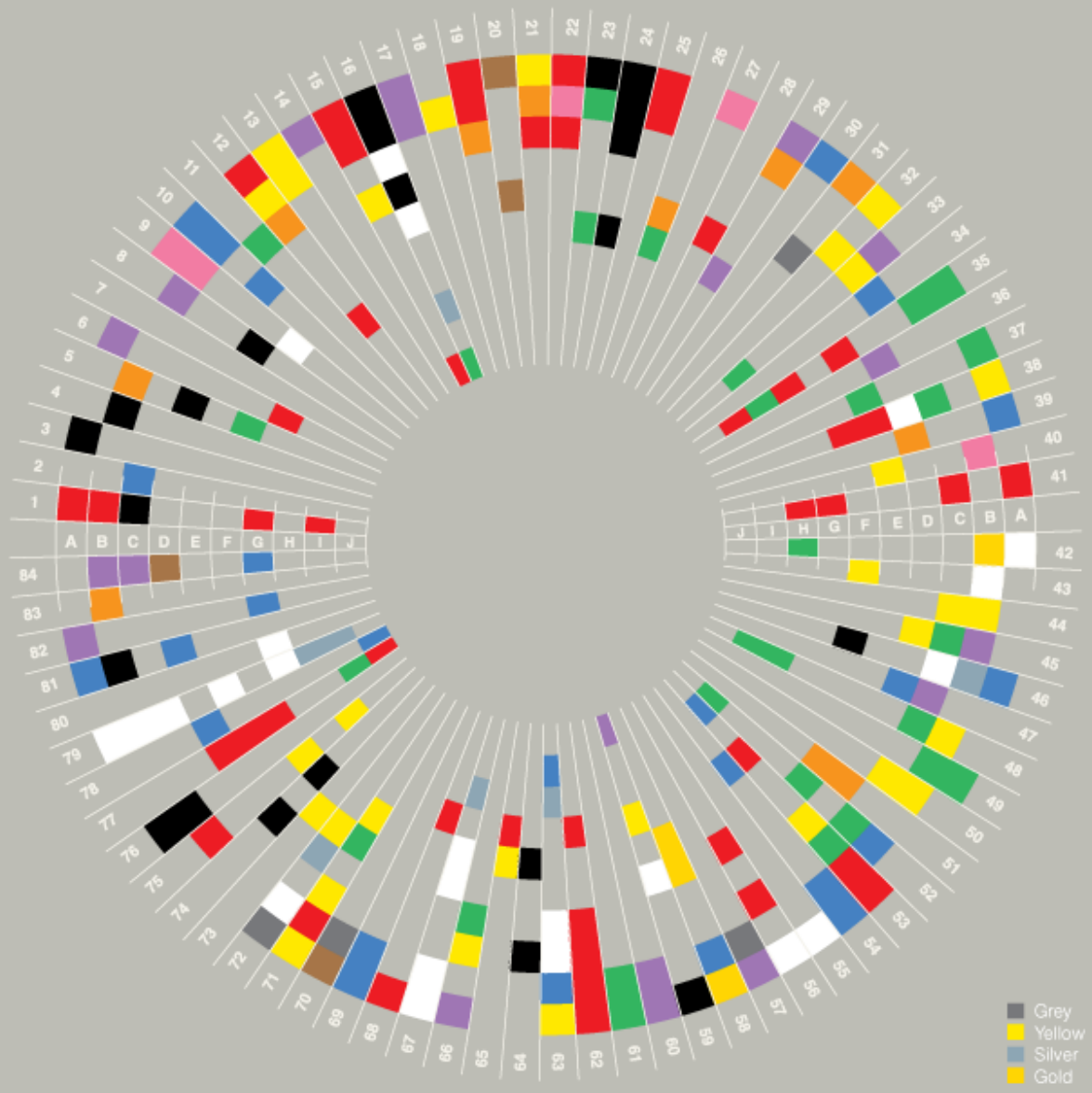
Slide courtesy Ed Hawkins, EGU 2016 Making Better Scientific Graphics

“Chromatic Vision Simulator” Kazunori Asada mobile phone app



Slide courtesy Ed Hawkins, EGU 2016 Making Better Scientific Graphics

Colours In Culture



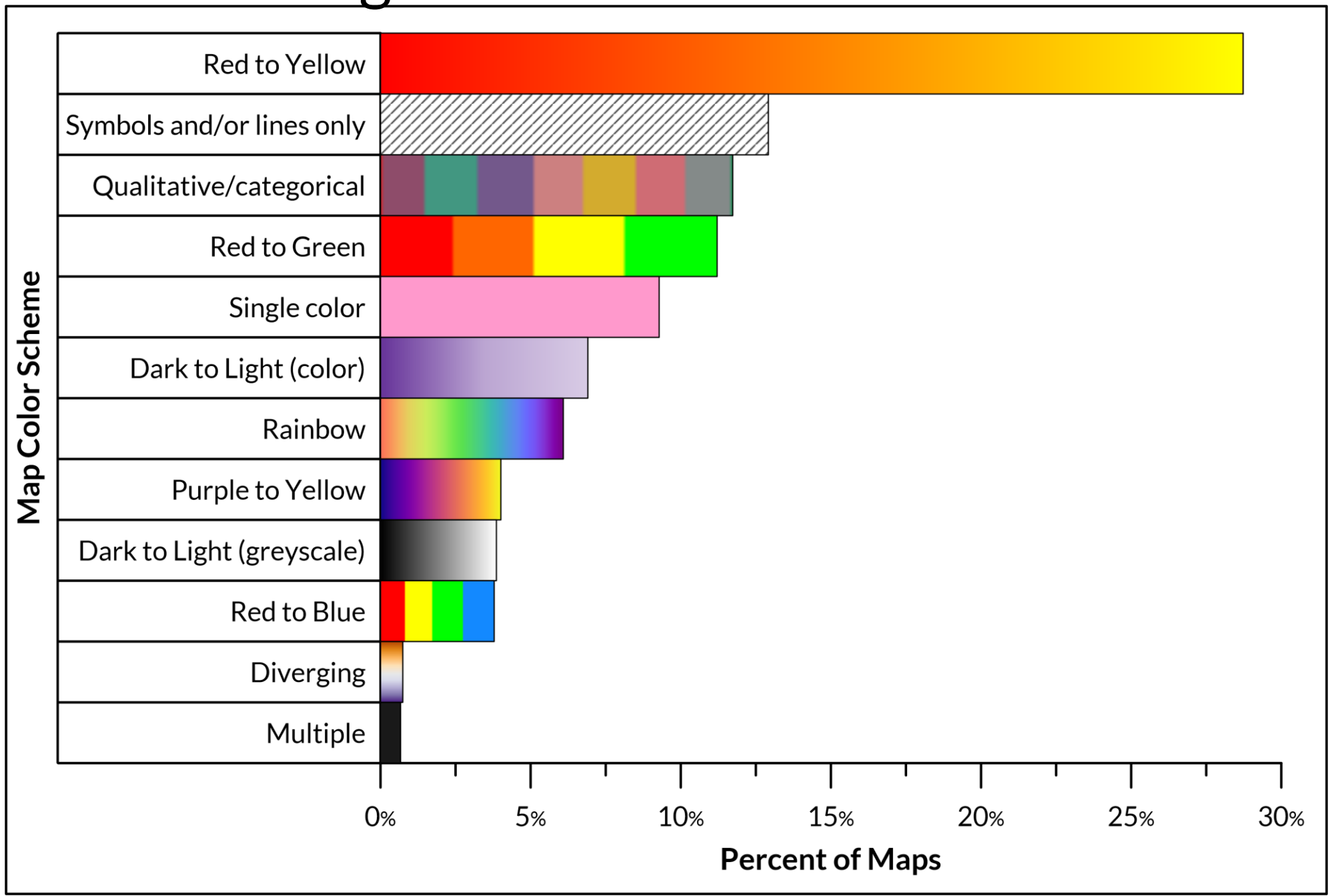
1 Anger
2 Art / Creativity
3 Authority
4 Bad Luck
5 Balance
6 Beauty
7 Calm
8 Celebration
9 Children
10 Cold
11 Compassion
12 Courage
13 Cowardice
14 Cruelty
15 Danger
16 Death
17 Decadence
18 Deceit
19 Desire
20 Earthy
21 Energy
22 Erotic
23 Eternity
24 Evil
25 Excitement
26 Family
27 Femininity
28 Fertility
29 Flamboyance
30 Freedom
31 Friendly
32 Fun
33 God
34 Gods
35 Good Luck
36 Gratitude
37 Growth
38 Happiness
39 Healing
40 Healthy
41 Heat
42 Heaven
43 Holiness
44 Illness
45 Insight
46 Intelligence
47 Intuition
48 Religion
49 Jealousy
50 Joy
51 Learning
52 Life
53 Love
54 Loyalty
55 Luxury
56 Marriage
57 Modesty
58 Money
59 Mourning
60 Mystery
61 Nature
62 Passion
63 Peace
64 Penance
65 Power
66 Personal power
67 Purity
68 Radicalism
69 Rational
70 Reliable
71 Repels Evil
72 Respect
73 Royalty
74 Self-cultivation
75 Strength
76 Style
77 Success
78 Trouble
79 Truce
80 Trust
81 Unhappiness
82 Virtue
83 Warmth
84 Wisdom

David McCandless & AlwaysWithHonor.com // v1.0 // Apr 09 // InformationIsBeautiful.net

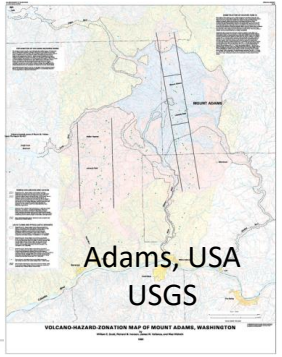
source: Pantone, ColorMatters & web sources

The risk hierarchy for color (red means danger, orange means warning, yellow means caution, and green mean safe) has found to be widespread among cultures [Wolgalter et al, 1997; Hupka et al, 1997; Dunlap at al, 1986]. Because of this though, red shouldn't be used for other things!

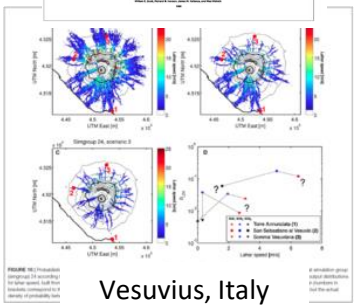
Design elements: color scheme



Design elements: Map publication format



Adams, USA
USGS



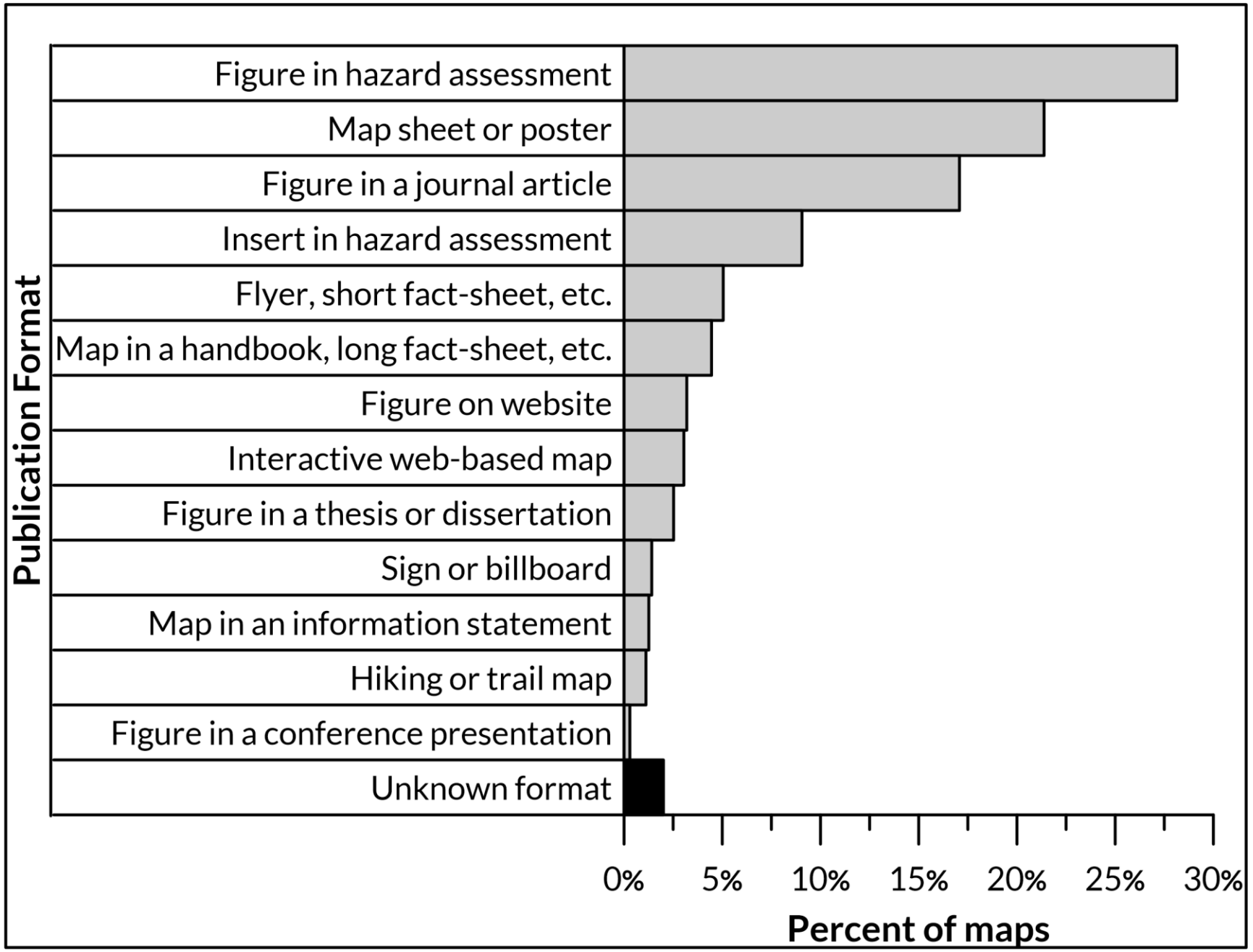
Vesuvius, Italy
Tierz et al. (2017)



Ambae, Vanuatu
Govt. Vanuatu



El Misti, Peru
INGEMMET

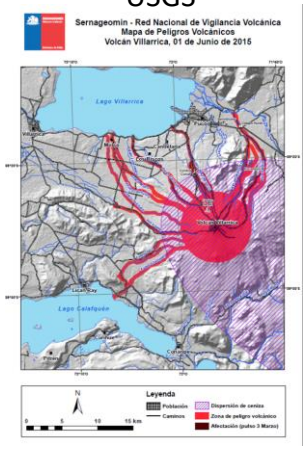




Bandai, Japan
Bandaisan Volcanic Disaster Management Council

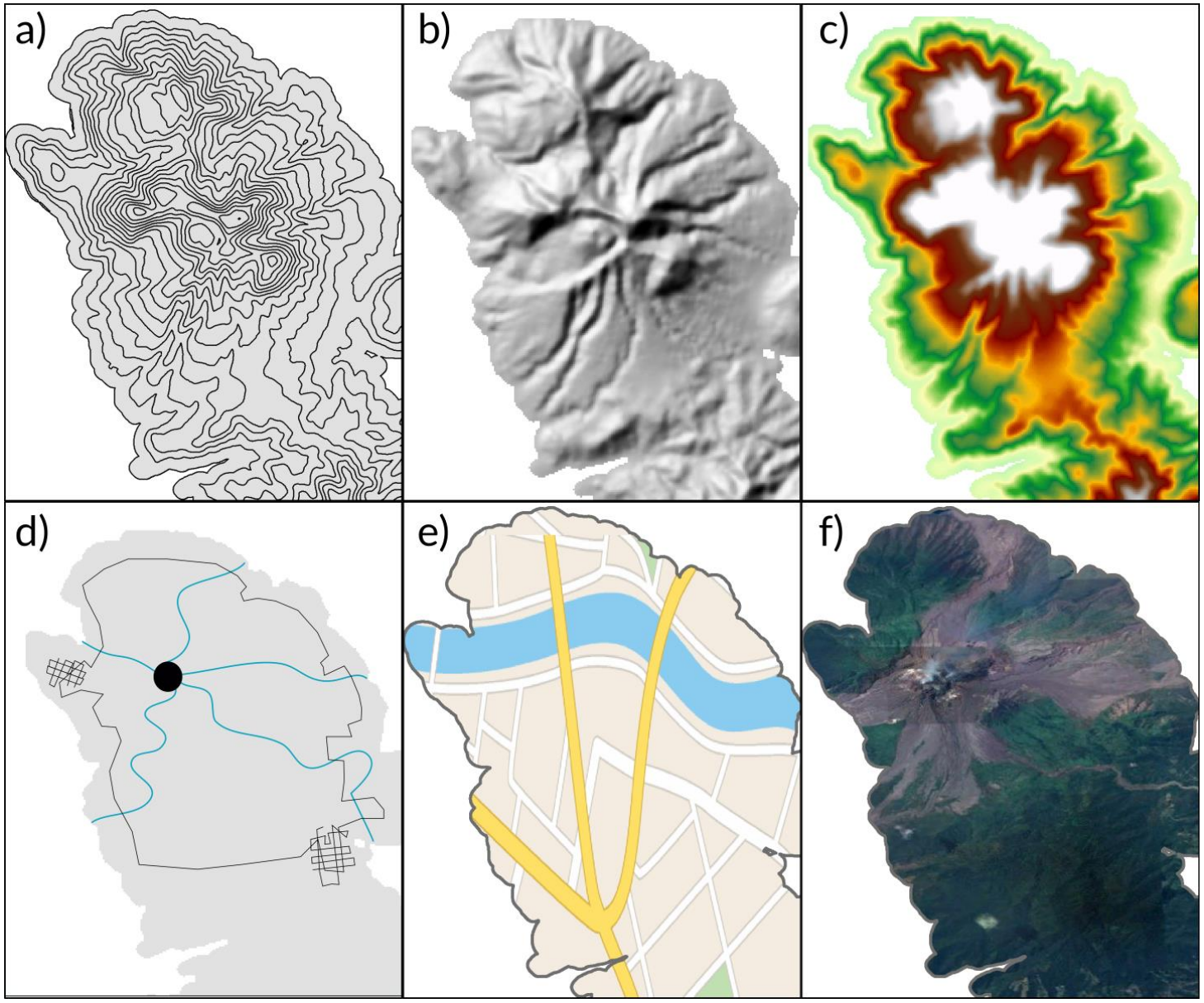


Rainier, USA
USGS



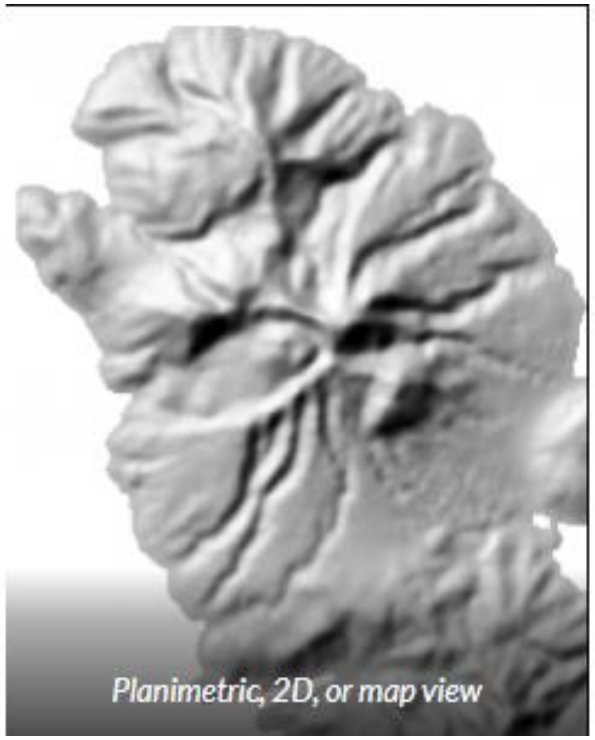
Villarrica, Chile
SERNAGEOMIN

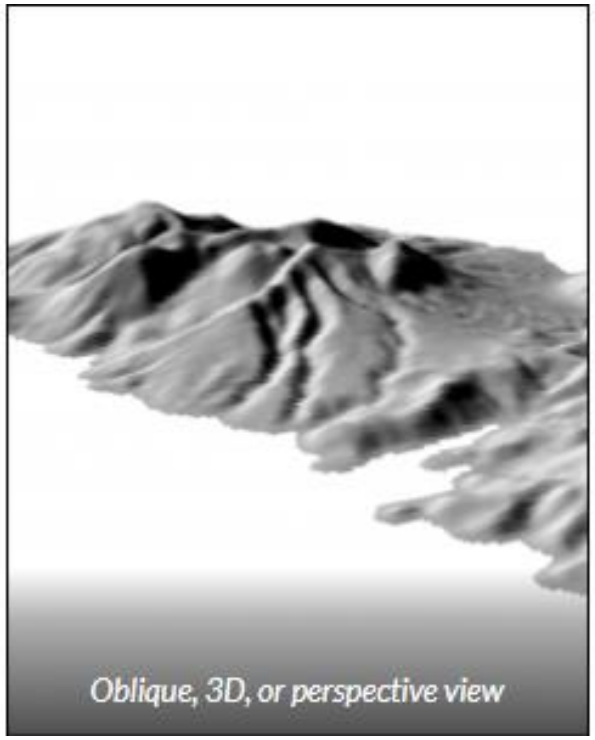
Design elements: basemaps and overlays

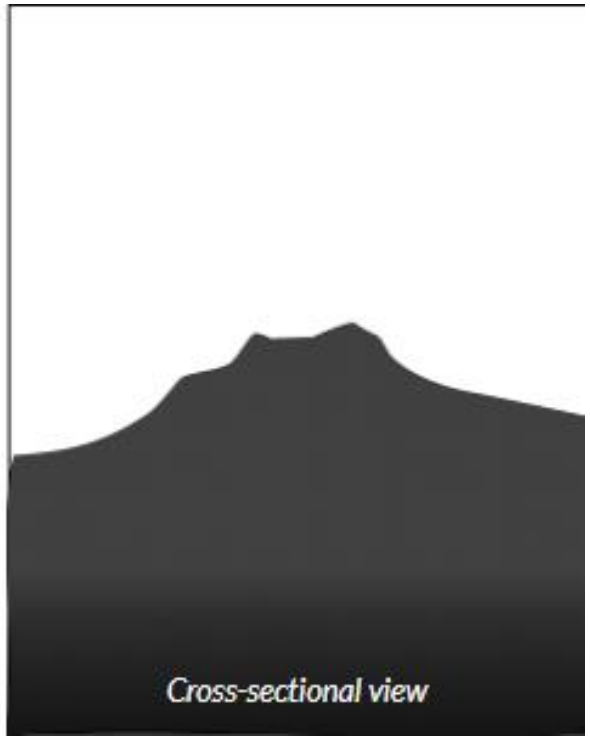


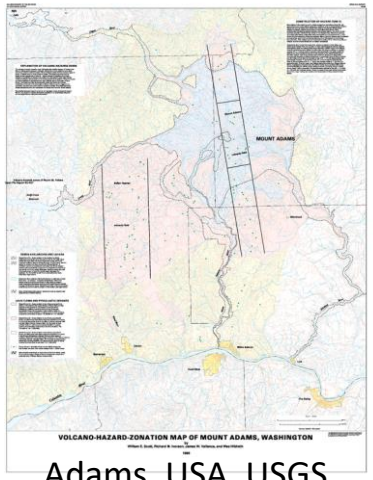
- Hillshade DEMs are the most common basemap used by volcanic hazard maps in the database (44%).
- Studies have shown that hillshade maps are much more intuitive for topography interpretation than contour maps (Haugerud & Greenberg 1998; Konnelly 2002). Contour maps require specialized training to interpret and are not intuitive for the general public
- Research has shown that satellite images or photographs are particularly easy for the general public to interpret (Haynes et al. 2007). However, it may be difficult to layer hazard zone and other information on photographs without the map becoming cluttered or obscuring the photographic base.

Design elements: map view/dimensionality





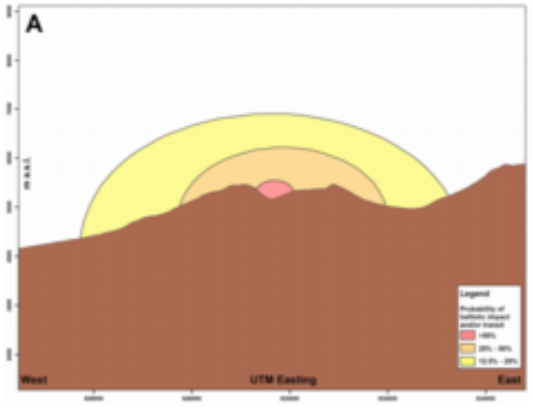




Adams, USA, USGS



El Misti, Peru
INGEMMET

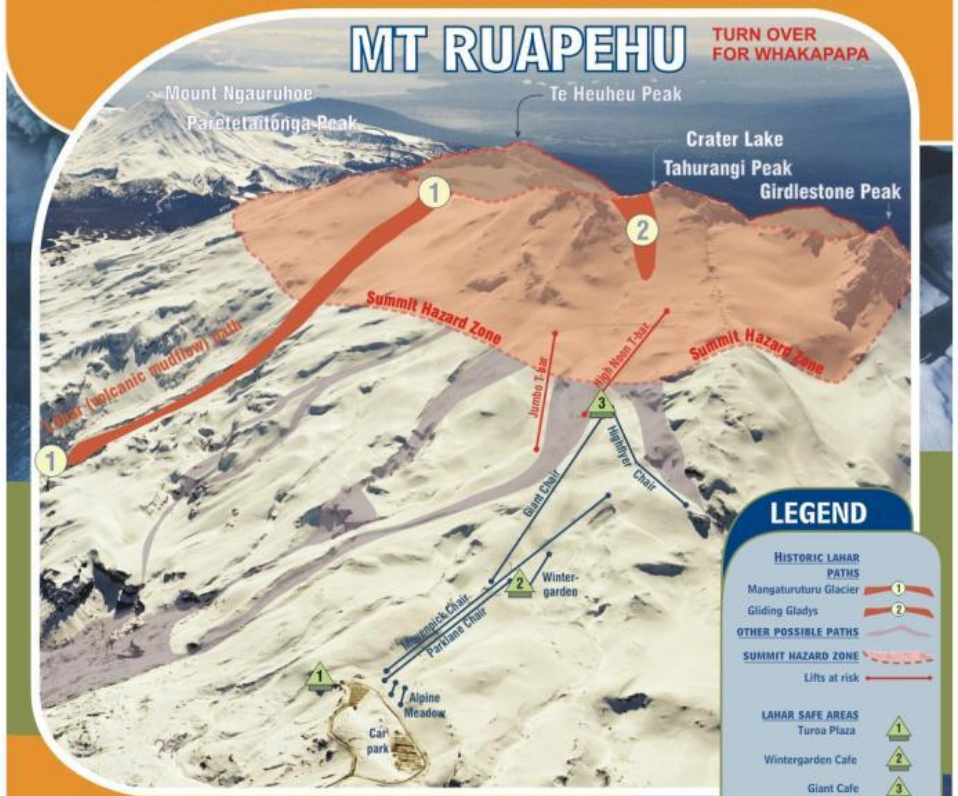


Lascar, Chile
Bertin (2017)

VOLCANIC HAZARDS AT TUROA

MT RUAPEHU

TURN OVER FOR WHAKAPAPA



LEGEND

- HISTORIC LAHAR PATHS
  - Mangatururu Glacier (1)
  - Gliding Gladys (2)
- OTHER POSSIBLE PATHS
- SUMMIT HAZARD ZONE
- Lifts at risk
- LAHAR SAFE AREAS
  - Turoa Plaza (1)
  - Wintergarden Cafe (2)
  - Giant Cafe (3)
- Safe lift lines

WHAT TO DO!!

If there are any signs of an eruption (earthquakes, rumbling from crater, ash cloud or flying rocks):

- Move as quickly as possible down the mountain from the two at-risk T-bars (Jumbo and High Noon) and all trails at their elevation.
- Stay on ridges, out of valleys as much as possible.
- Below the at-risk T-bars (Jumbo and High Noon) move out of the valleys by removing your skis or board and climbing the valley walls to a ridge top.
- Know where the safe areas are (buildings, lift lines and ridges below Jumbo and High Noon T-bars)
- Stay in safe areas until advised otherwise by ski area staff.

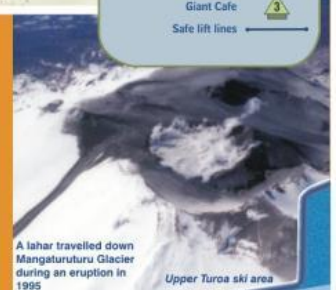
HAZARDS

SUMMIT HAZARDS

- Ruapehu is an active volcano
- During an eruption there may be gas and flying rocks on the upper mountain, especially within the Summit Hazard Zone
- This zone includes upper Jumbo and High Noon T-Bars

LAHARS

- Eruptions generate lahars (volcanic mudflows) which move down valleys in a flash flood.
- During an eruptions stay out of valleys and follow instructions from staff.



A lahar travelled down Mangatururu Glacier during an eruption in 1995

Upper Turoa ski area

GEOLOGICAL & NUCLEAR SCIENCES

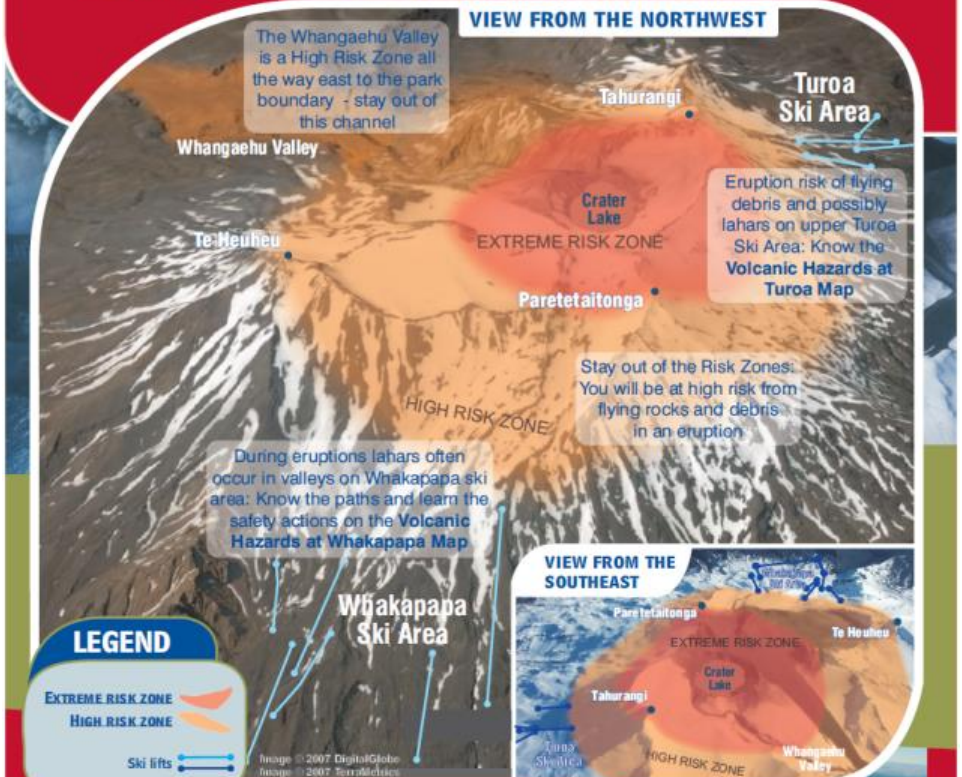
Department of Conservation Te Papa Atawbai

MT RUAPEHU WHAKAPAPA TUROA NGA HAUHUNGA

Disclaimer
Volcanic activity and other unforseen factors may alter features on this map, so safe and hazardous areas may change without notice.

ERUPTION HAZARDS AT MT RUAPEHU

VIEW FROM THE NORTHWEST



VIEW FROM THE SOUTHEAST

LEGEND

- EXTREME RISK ZONE
- HIGH RISK ZONE
- Ski lifts

WHAT TO DO!!

Stay out of High and Extreme Risk Zones

If there are any signs of an eruption (earthquakes, rumbling from crater, ash-steam cloud or flying rocks):

- Move as quickly as possible down the mountain away from Risk Zones.
- Stay on ridges, out of valleys as much as possible.
- Move out of the valleys by removing your skis or board and climbing the valley walls to a ridge line.
- Know where the safe areas are (buildings and ridges below the Risk Zones)
- Stay in safe areas until advised by ski area or search and rescue personnel.

HAZARDS

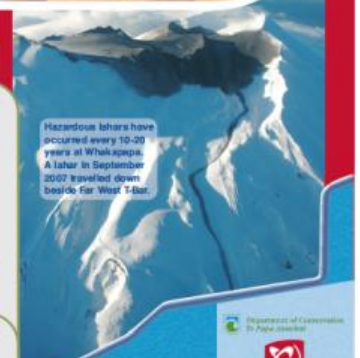
Ruapehu is an active volcano.

- During an eruption there will be flying rocks on the upper mountain, especially within the Extreme and High Risk Zones.
- Eruptions generate lahars (volcanic mudflows), which flow down valleys in a flash flood.

WARNING SYSTEM FOR SKI AREA

An Eruption Detection System (EDS) is operating at Whakapapa Ski Area.

- The system will set off sirens and loudspeaker messages.
- Follow instructions from staff and move out of valleys.



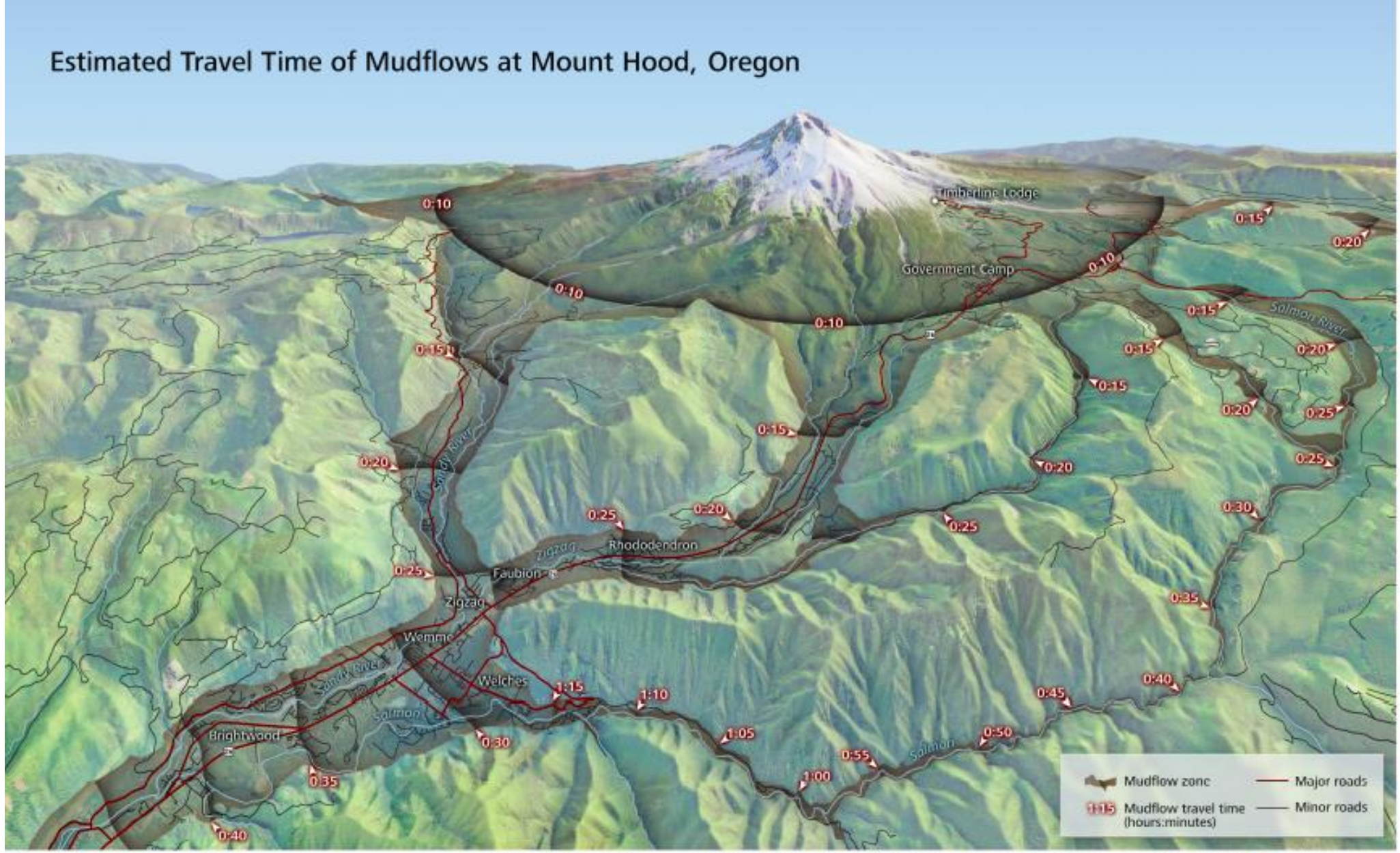
Department of Conservation Te Papa Atawbai

GNS SCIENCE

MT RUAPEHU WHAKAPAPA TUROA NGA HAUHUNGA

Published: 8 October 2007
Disclaimer: Volcanic activity and other unforseen factors may alter features on this map, so safe and hazardous areas may change without notice.

Estimated Travel Time of Mudflows at Mount Hood, Oregon





Terrain: National Elevation Dataset
Imagery: National Agriculture Imagery Program
Roads: OpenStreetMap.org via Geofabrik
Raw mudflow data and general information on Mount Hood: United States Geological Survey

Charles Pregpermau and Bernhard Jenny, Oregon State University, 2015
© 2015 Journal of Maps

The author of this map is not affiliated with the United States Geological Survey. For official information on volcanic hazards in the United States, visit http://volcanoes.usgs.gov/. This map is for research and display purposes only; it is not intended or recommended for use in emergency planning or any other practical application. Conclusions drawn from the information in this map are the sole responsibility of the user.

Oregon State University OSU
CARTOGRAPHY & GEOVISUALIZATION
http://cartography.oregonstate.edu/

Volcano Fact Sheet

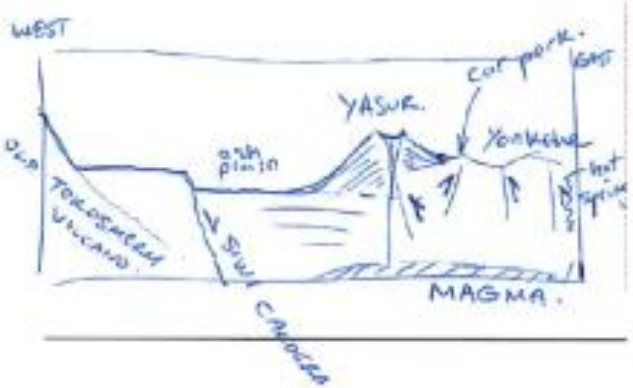
Yasur Volcano – Yenkahe Caldera



Description

- Yasur, located at the SE tip of Tanna Island, is a mostly unvegetated 361-m-high scoria cone with a nearly circular, 400-m-wide summit crater.
- Yasur is the most frequently visited of the Vanuatu volcanoes.
- It has been in more-or-less continuous activity since Captain Cook observed ash eruptions in 1774.
- This style of activity may have continued for the past 800 years.

(www.volcano.si.edu)

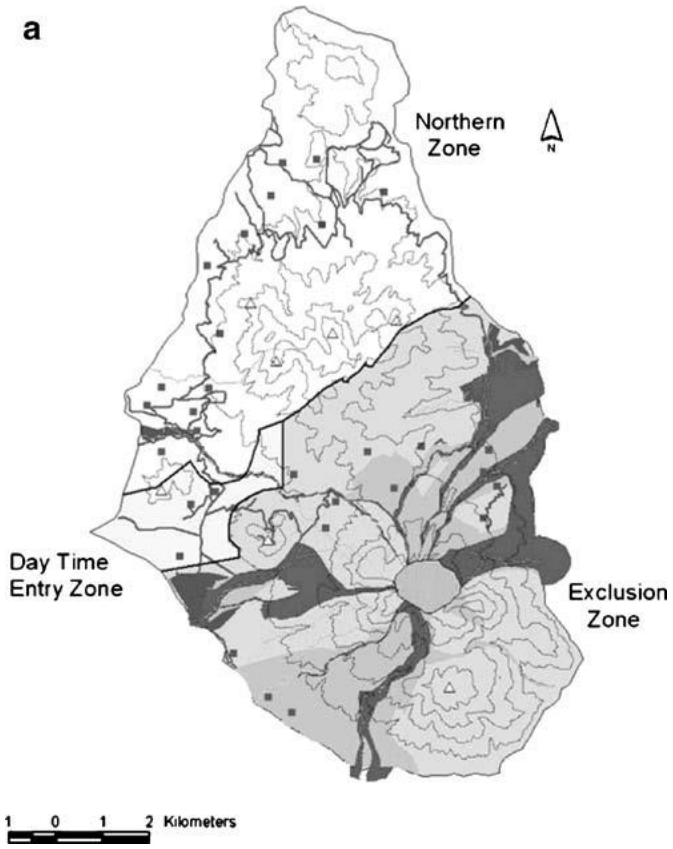


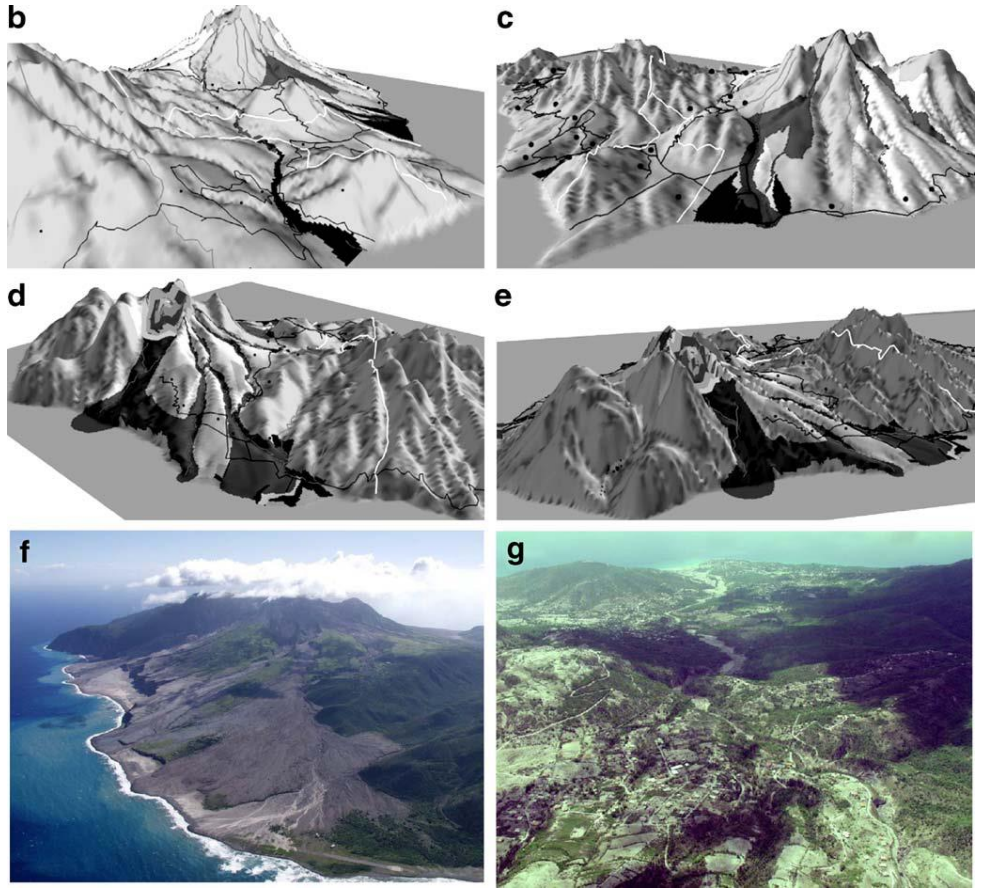


Dimensionality and base-maps

- Each of these has been shown to have particular advantages and disadvantages that may vary cross-culturally [Bostrom et al, 2008].

- In general, 3D visualizations are better at conveying shape, while 2D visualizations facilitate the understanding of the relative position of two objects [Bostrom et al, 2008].

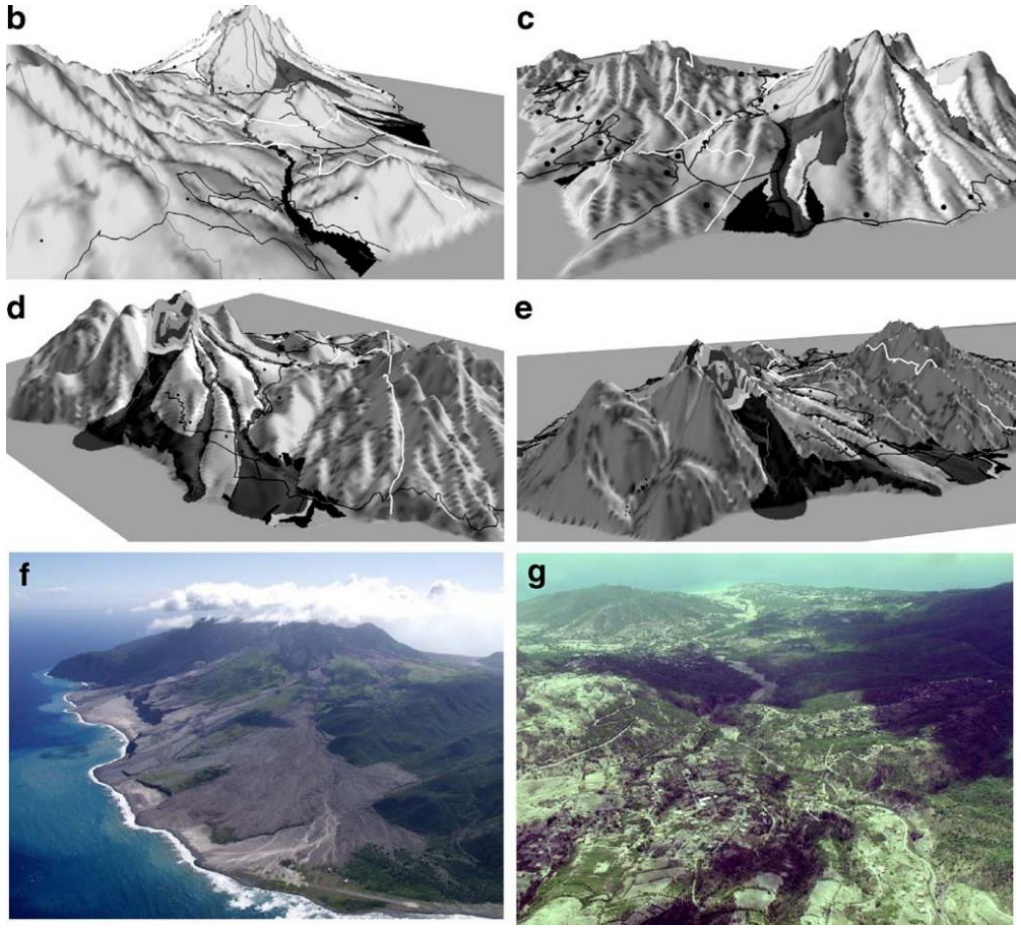




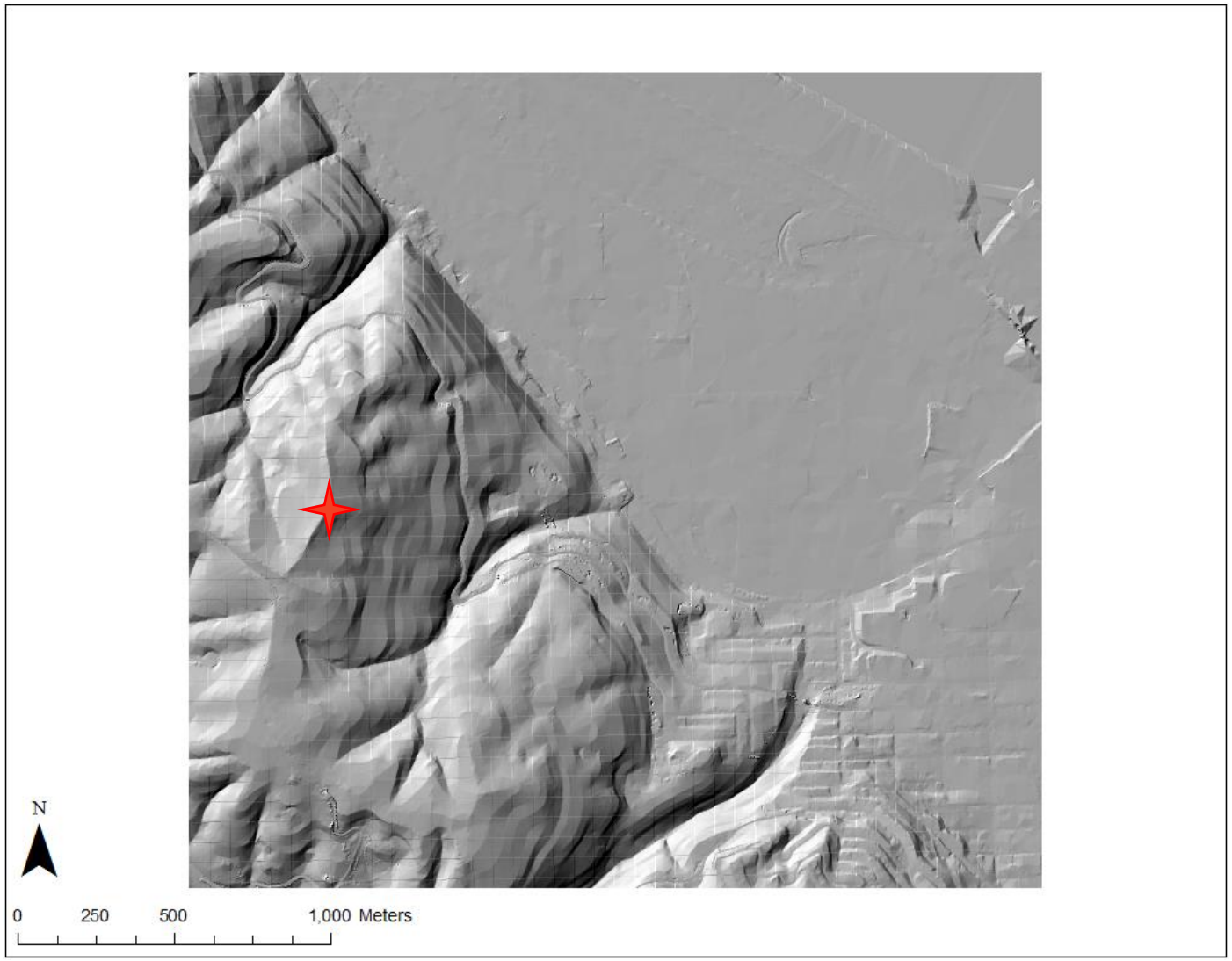
Haynes, K., Barclay, J., and Pidgeon, N. 2007. Volcanic hazard communication using maps: an evaluation of their effectiveness. Bulletin of Volcanology, 70: 123-138.

Dimensionality and base-maps

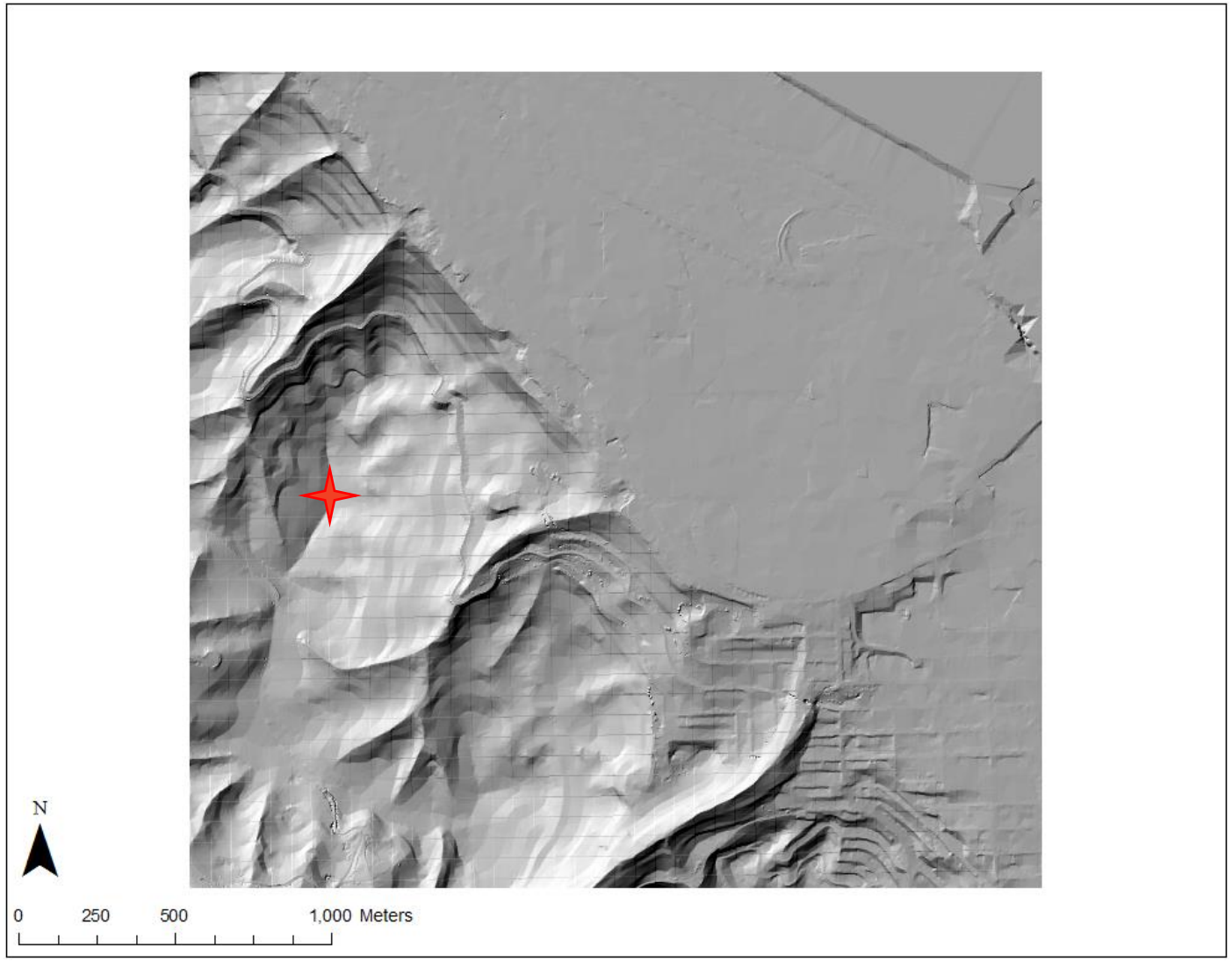
- Orienting: Finding north was easier on the 2-D contour map, perhaps because maps are normally oriented with north at the top, otherwise similar
- Locating: easier using photographs than 3D maps or 2D contour.
- Map skills: Good when using photographs and 3D maps. Contour lines were found to be extremely confusing. Often, linear features such as roads, the boundaries of the exclusion zone, and the contours themselves were confused with one another.
- Volcanic information: improved using 3D and photographic maps over 2D contour maps. Some respondents reevaluated their perceptions of danger when they were presented with the aerial photograph maps. One woman was shocked to discover that the Belham Valley was a direct pathway for pyroclastic flows to travel from the volcano, whereas before she couldn't understand why the area had been evacuated.



Hillshade

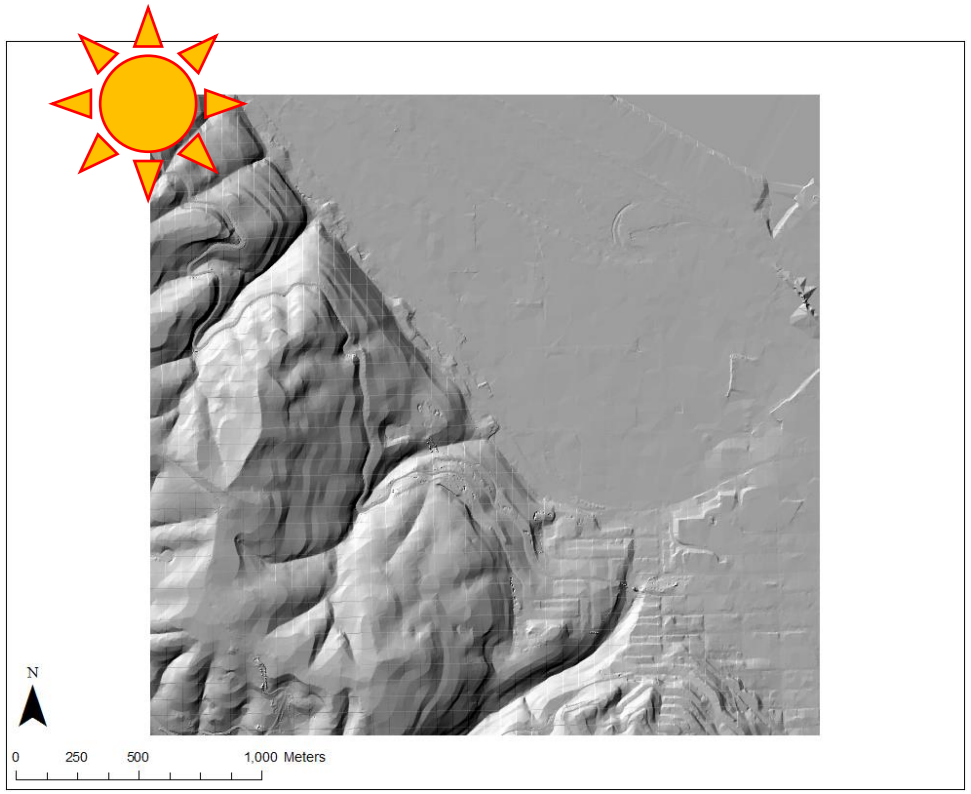


Hillshade

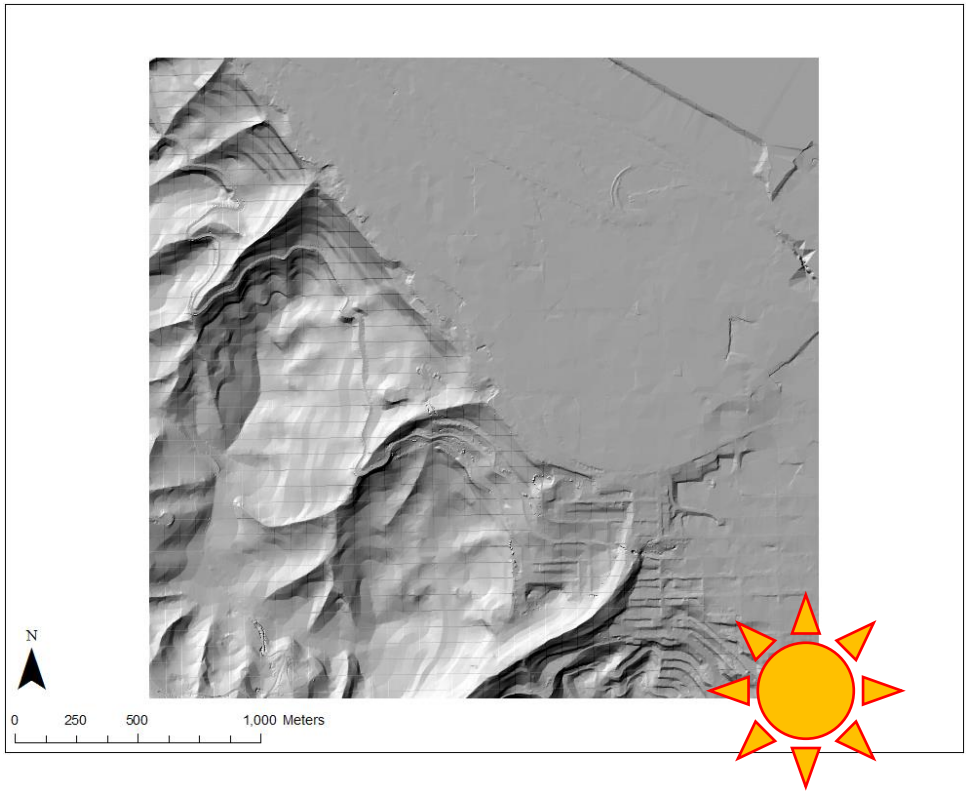


Hillshade

Illuminate from top left (from NW, 315°)

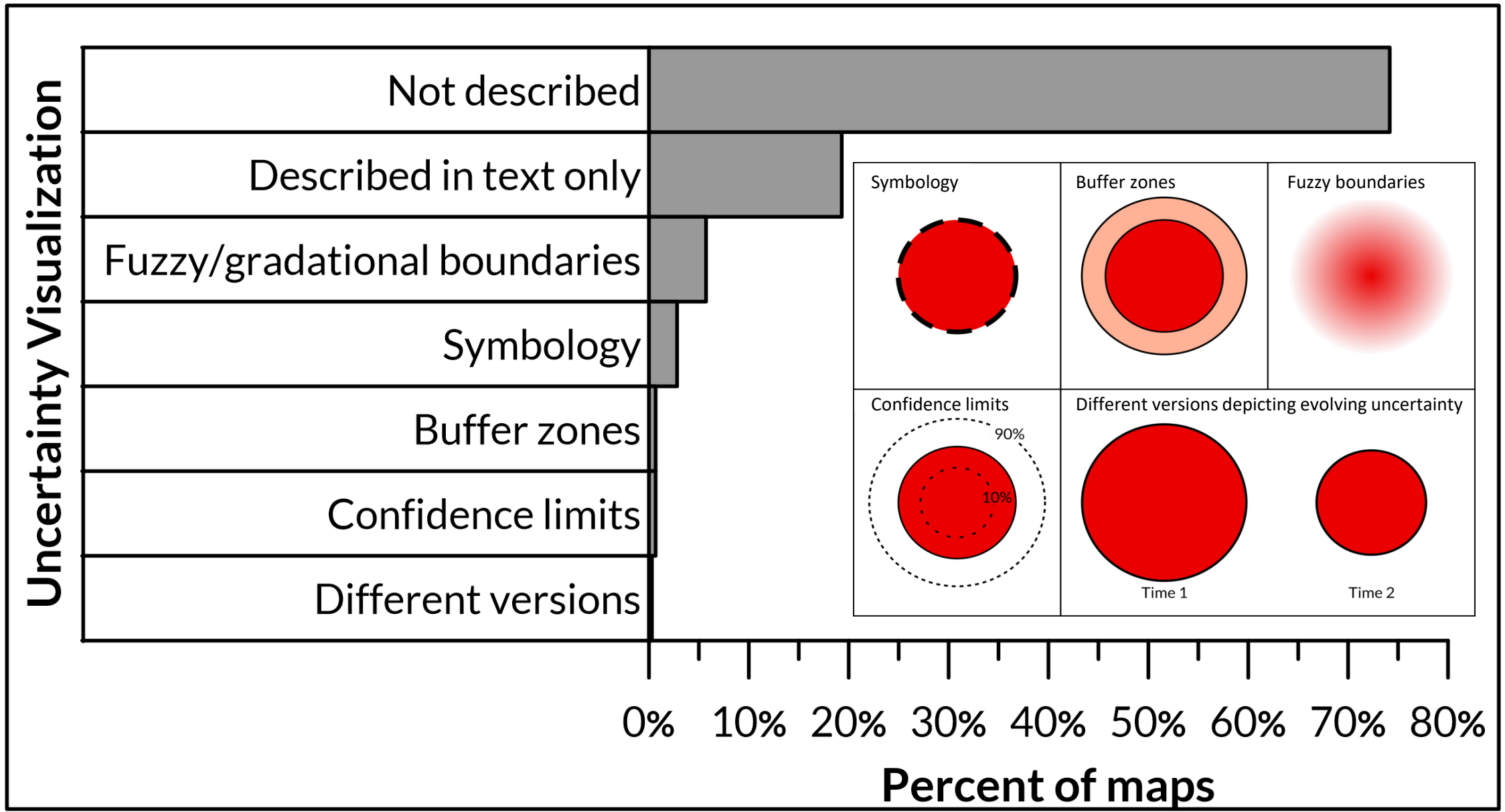


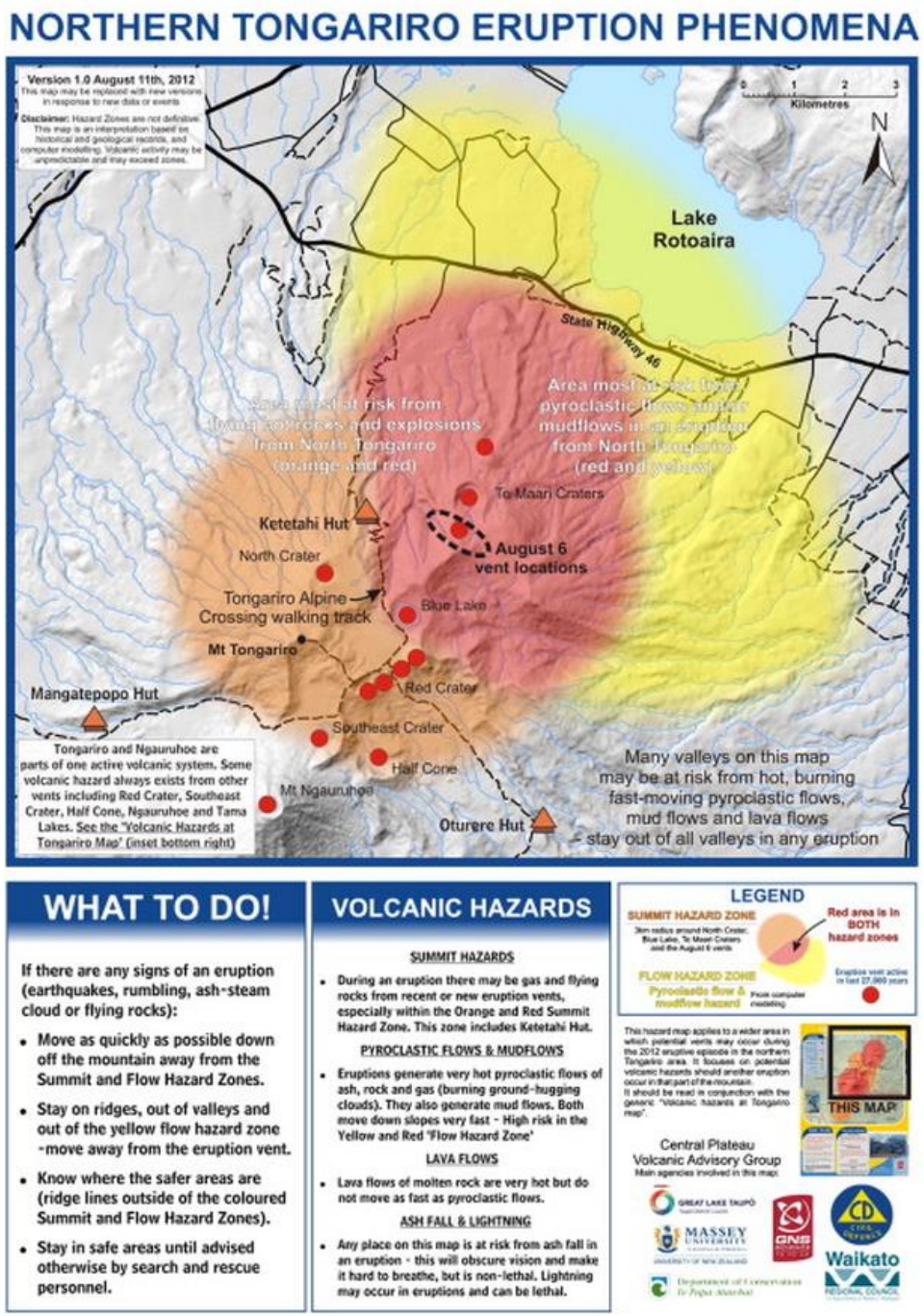
Illuminate from bottom right (from SE)



This is known as the **pseudoscopic illusion**, and cartographers have known that it impacts the interpretation of shaded relief maps for a very long time. **While most people will interpret terrain correctly when shaded from a northwest illuminator, there is a smaller (but not insignificant) group of people that will actually perceive these maps in completely the opposite way**.

Hazard Zone Uncertainty

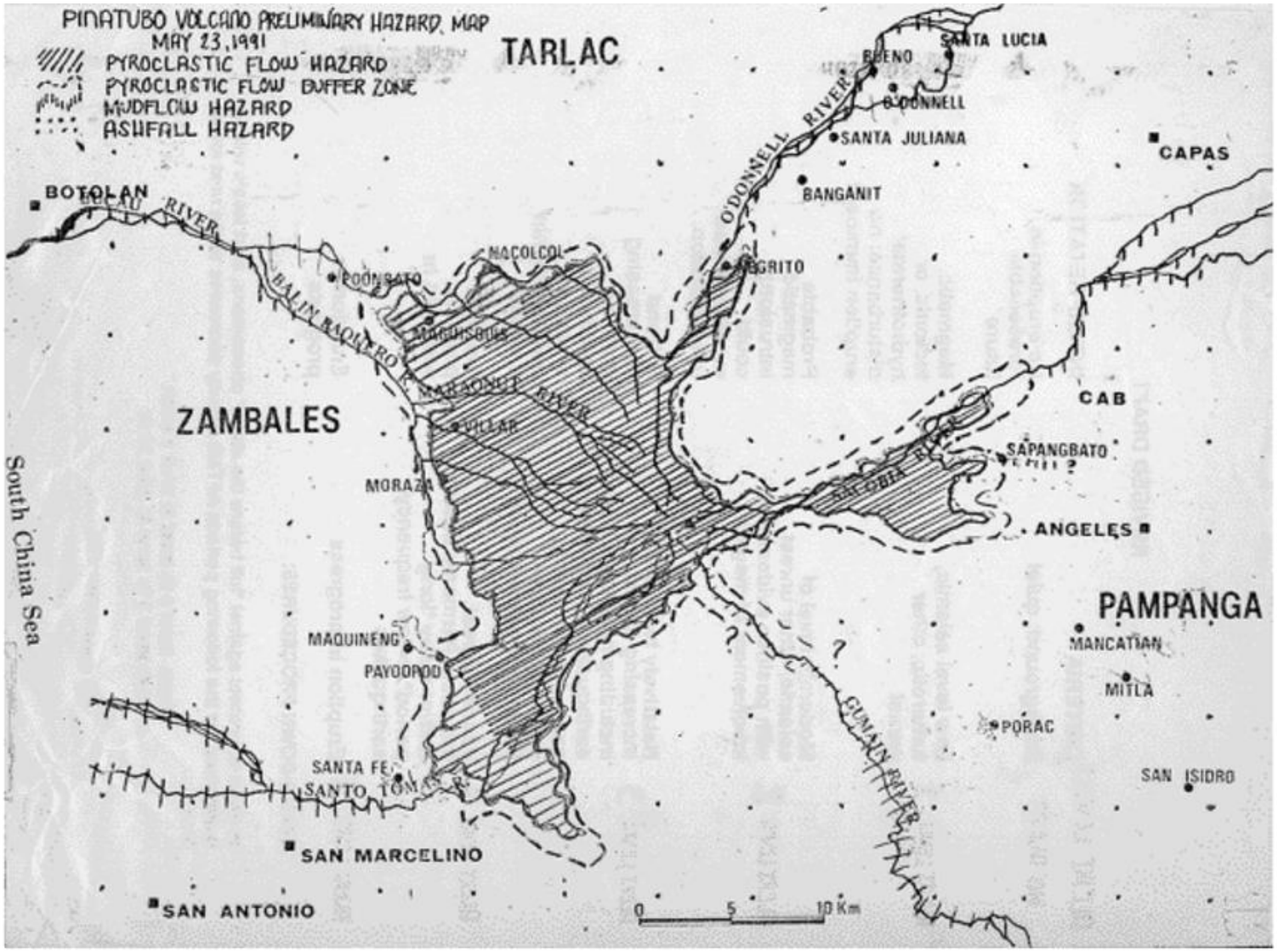




Tongariro, New Zealand (GNS, 2012)

Fuzzy boundaries





Buffer zones

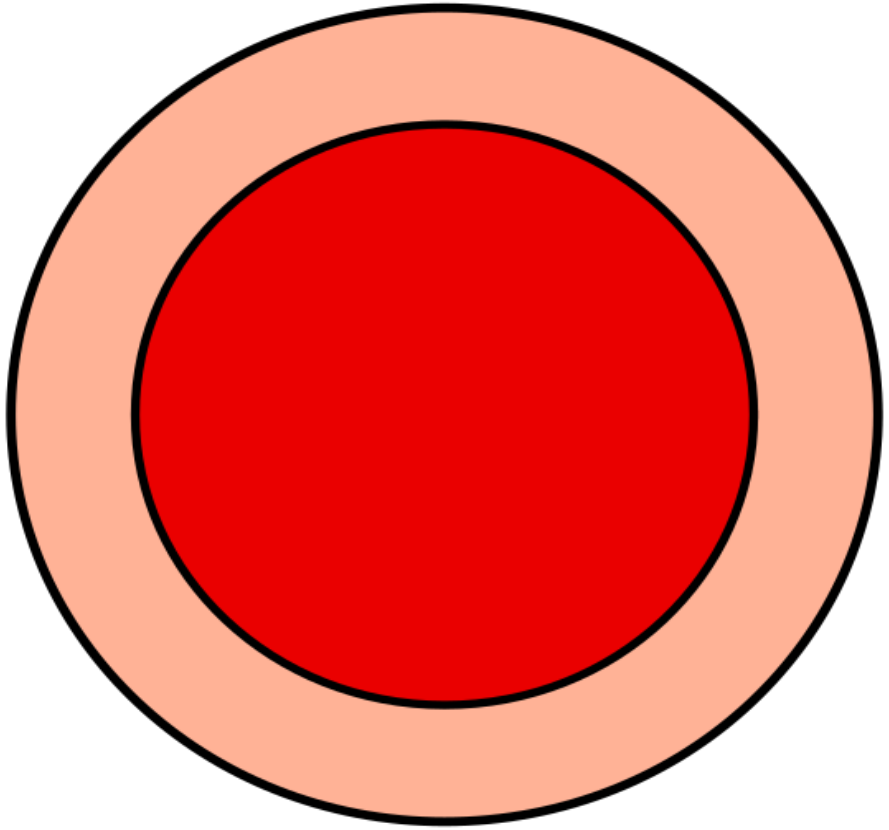


Figure 5. Hazard map distributed to provincial authorities beginning on May 23, 1991. All three major hazards-pyroclastic flows, ash fall, and lahars-are shown on this single map. Ash-fall hazard is not adjusted for prevailing winds. Question marks along the Gumain River reflect uncertainty about whether pyroclastic flows had jumped a high ridge near Mount Pinatubo to enter the drainage.

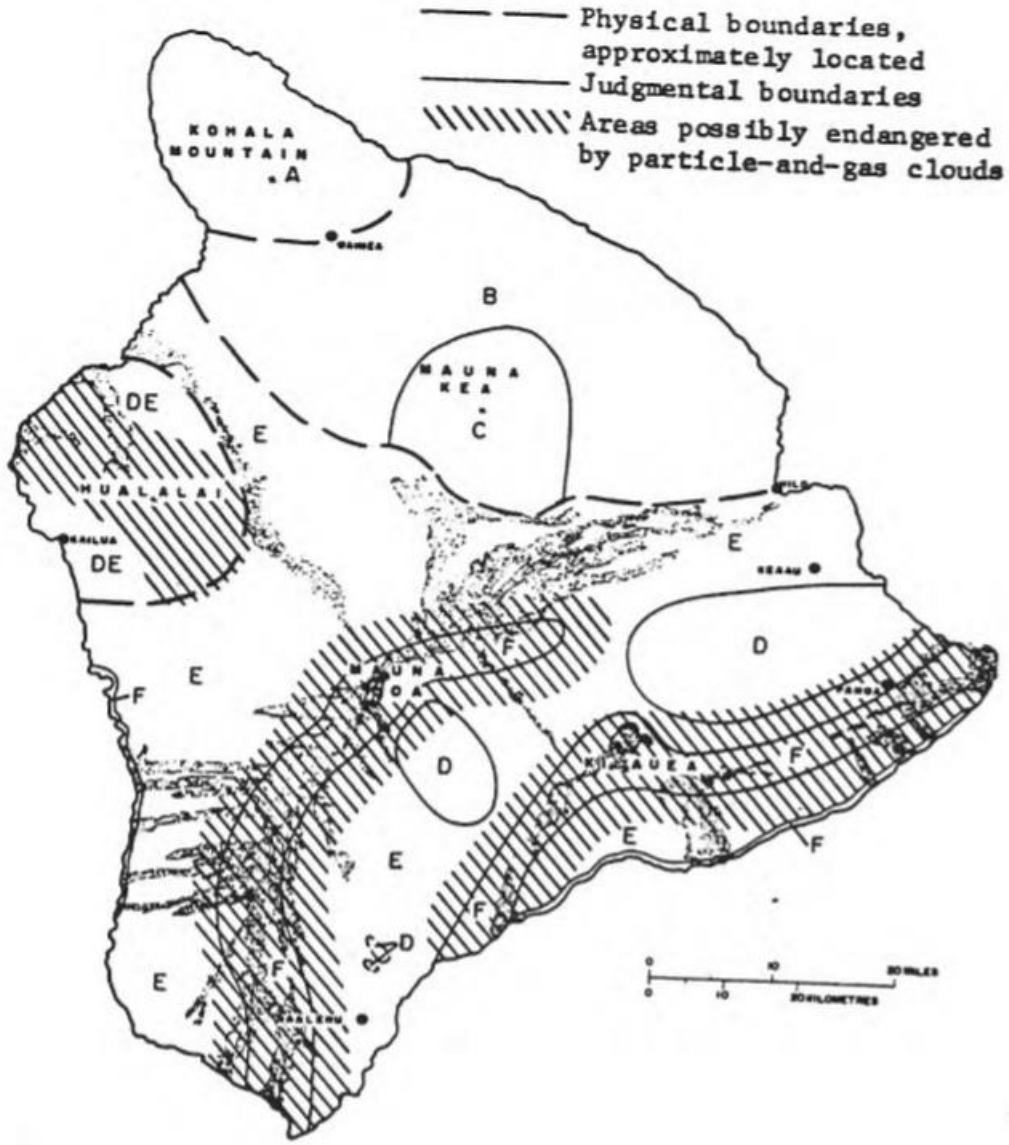
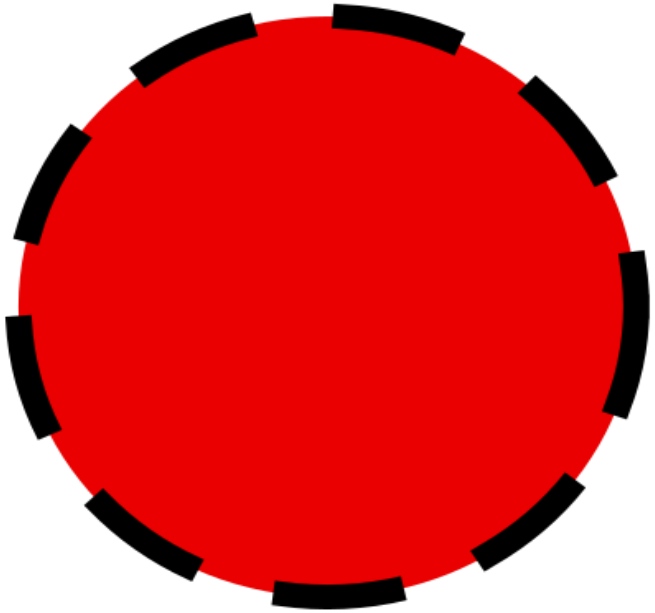


Figure 15.--Zones of overall relative risk from volcanic hazards. Risk increases from "A" through "F".

Symbology



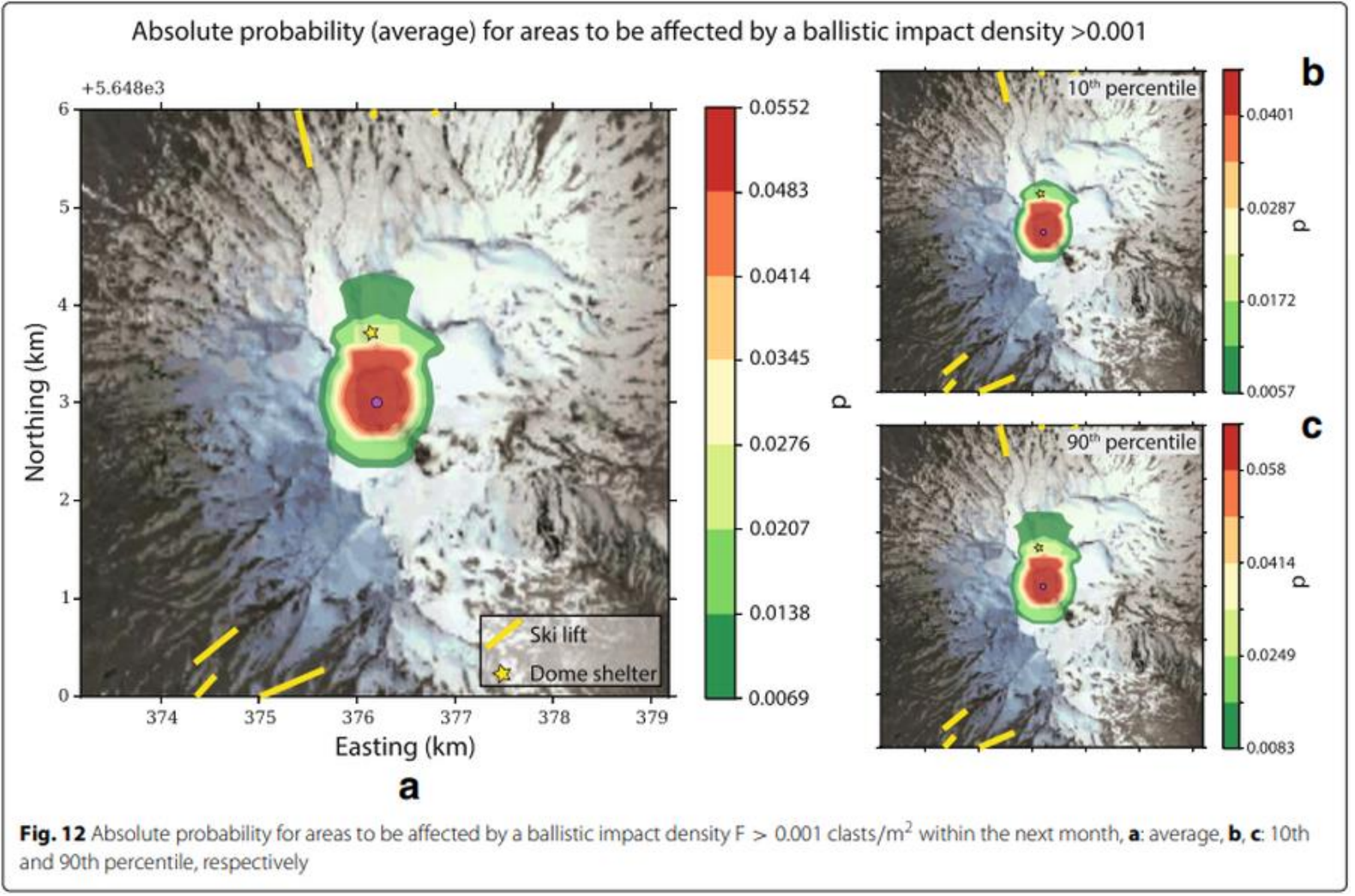
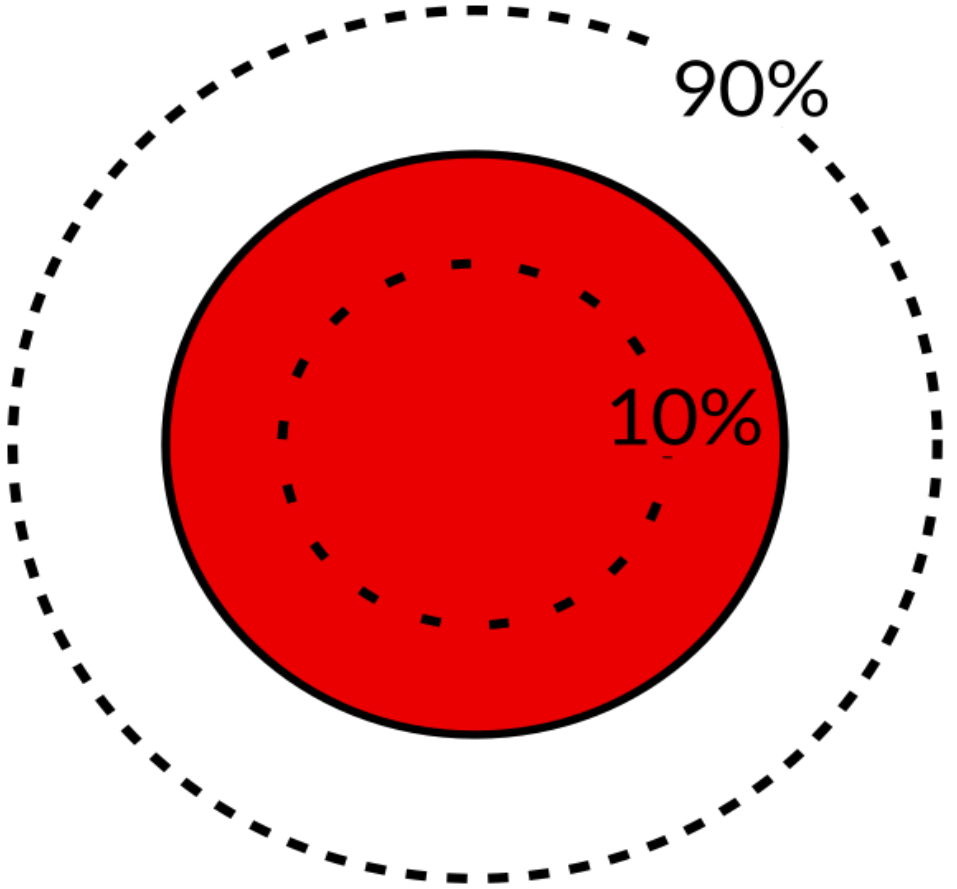


Fig. 12 Absolute probability for areas to be affected by a ballistic impact density F > 0.001 clasts/m^2 within the next month, **a**: average, **b**, **c**: 10th and 90th percentile, respectively

Confidence intervals



Ruapehu, New Zealand (Strehlow et al. 2017)

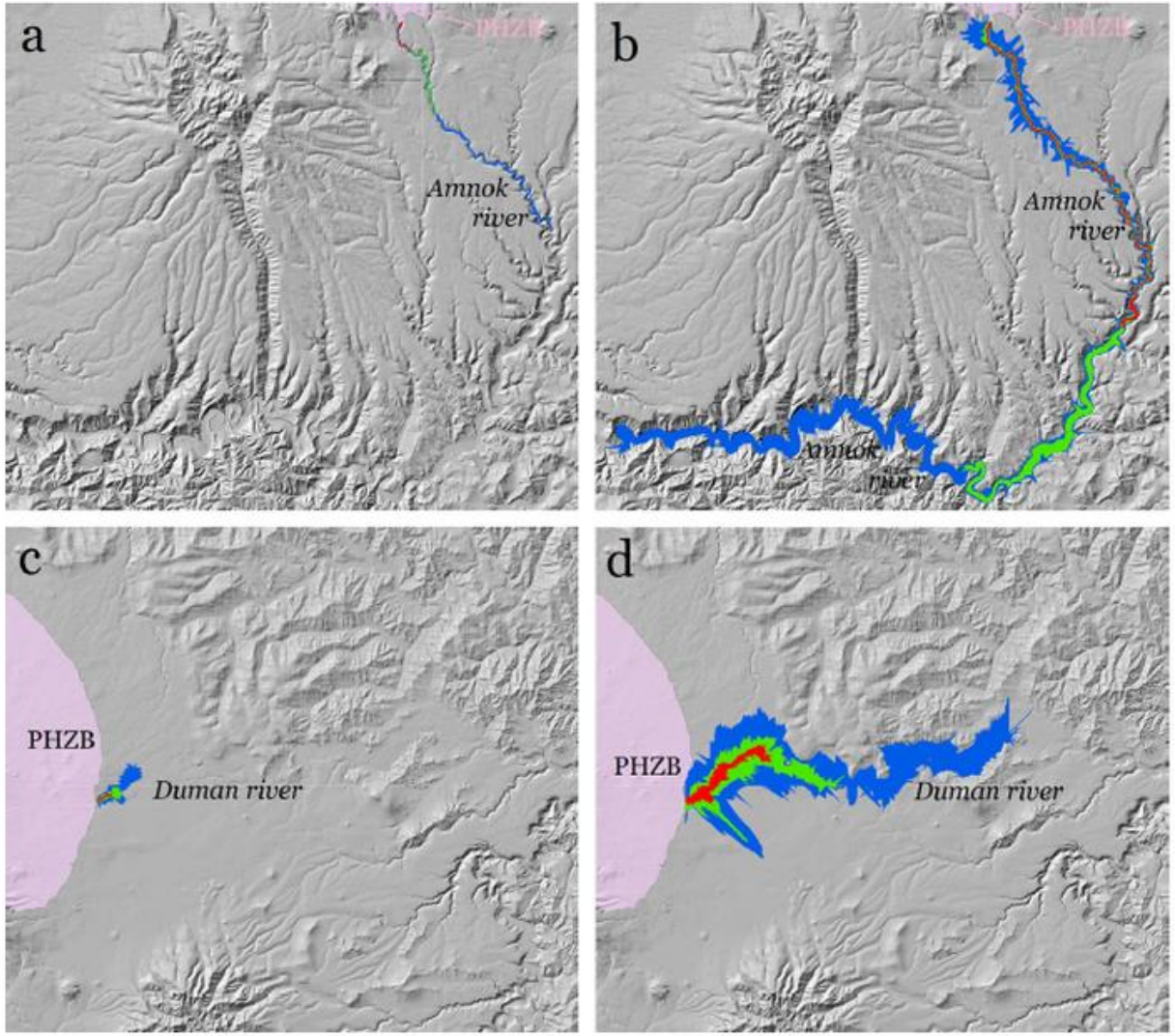
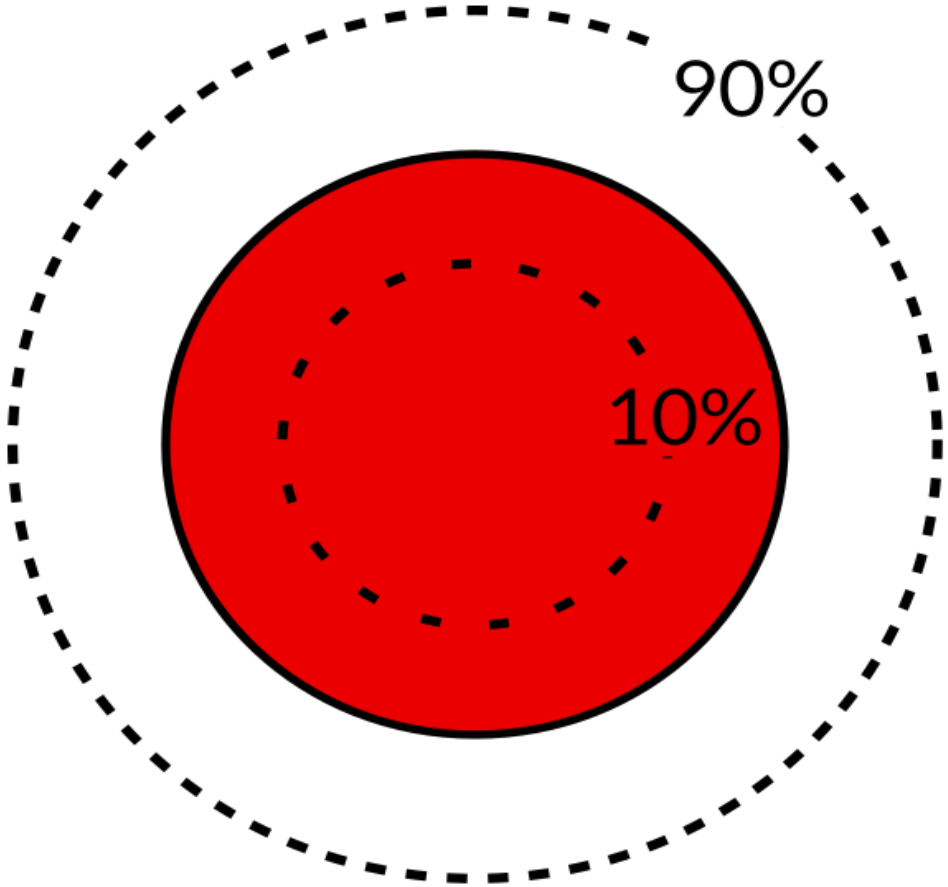


Fig. 5. Inundation areas of lahar at the Amnok river (a & b) and Duman river (c & d) on the hillshade images showing the probable range of planimetric area values with 95% confidence limits(green is the mean value from the regression line, blue is the upper confidence limit, red is the lower confidence limit) from 1×10^{6} m^3 (left) and 1×10^{8} m^3 (right) of lahar volumes.

Confidence intervals



Changbaishan [Baekdusan], China-North Korea (Yun & Chang, 2017)

Map element	Description	% of maps
actions to take	The map describes what to do during unrest or eruption	22
alert level scheme	The alert level scheme for the volcano is described	11
area map	An area map is included that shows the regional context	24
audience and/or purpose	The intended audience or purpose is described	11
cartographic legend	A legend is provided for cartographic symbols (e.g. roads, lakes)	52
color scheme order	Color scheme is in order (e.g. red = high hazard)	58
conditional validity	The conditions under which the map is valid are described (e.g. VEI < 5, central vent eruptions)	26
coordinates	Geographic coordinates are shown	58
eruptive history	The eruptive history of the volcano is described	21
evacuation route	Evacuation routes are visually depicted on the map	10
expiration	The conditions that will trigger map revision are described (e.g. summit changes, a time limit, new information)	6
glossary of terms	Geological terms are defined in a glossary section	6
hazard details	Hazard processes are defined or described	18
hazard travel time	Hazard arrival times or velocities are depicted or described	13
hazard zone description	Hazard zones have accompanying descriptions beyond labels	74
hazard zone legend	A legend is provided for the hazard zones	90
impact details	Impact details (e.g. roof collapse, crop damage) are described	20
impact locations	Specific towns or drainages are named or listed	15
insets	Insets containing non-map information are included	26
methods	Hazard zonation methods are described in the text	22
more information source	Sources for additional information are given	17
north arrow	A north arrow is included	67
oblique image included	Oblique (3D) inset images of the terrain or hazard zones are shown	4
other volcanos	Hazard zones from nearby volcanos are also shown on the map	3
past deposits	Deposits from previous eruptions are shown visually	16
person hours	The amount of time required to make the map is stated	1
photos	Photos of the volcano, deposits, impacts, etc. are shown	12
population information	Population information (e.g. numbers of inhabitants in towns or hazard zones) is depicted or described	9
references	Reference literature is cited	16
safe areas	Safe areas, including shelters or muster points, are depicted	13
scale bar	A scale bar is included	86
version number	Version numbers or a revision history is provided	10
wind rose diagram	A wind rose diagram of either wind directions or tephra dispersal directions is provided	1

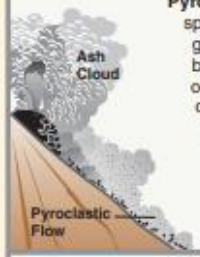
Are You at Risk from the Next Eruption of Mount Hood?

Volcano Hazards

Lava Flows and Domes

Lava is molten rock that flows onto the Earth's surface. **Lava flows** move downslope away from a vent and bury or burn everything in their paths. **Lava domes** form when lava piles up over a vent.

Pyroclastic Flows



Pyroclastic flows are high-speed avalanches of hot rock, gas, and ash that are formed by the collapse of lava domes or eruption columns. They can move up to 100 miles per hour and have temperatures to 1500°F. They are lethal, burning, burying, or asphyxiating all in their paths.

Tephra



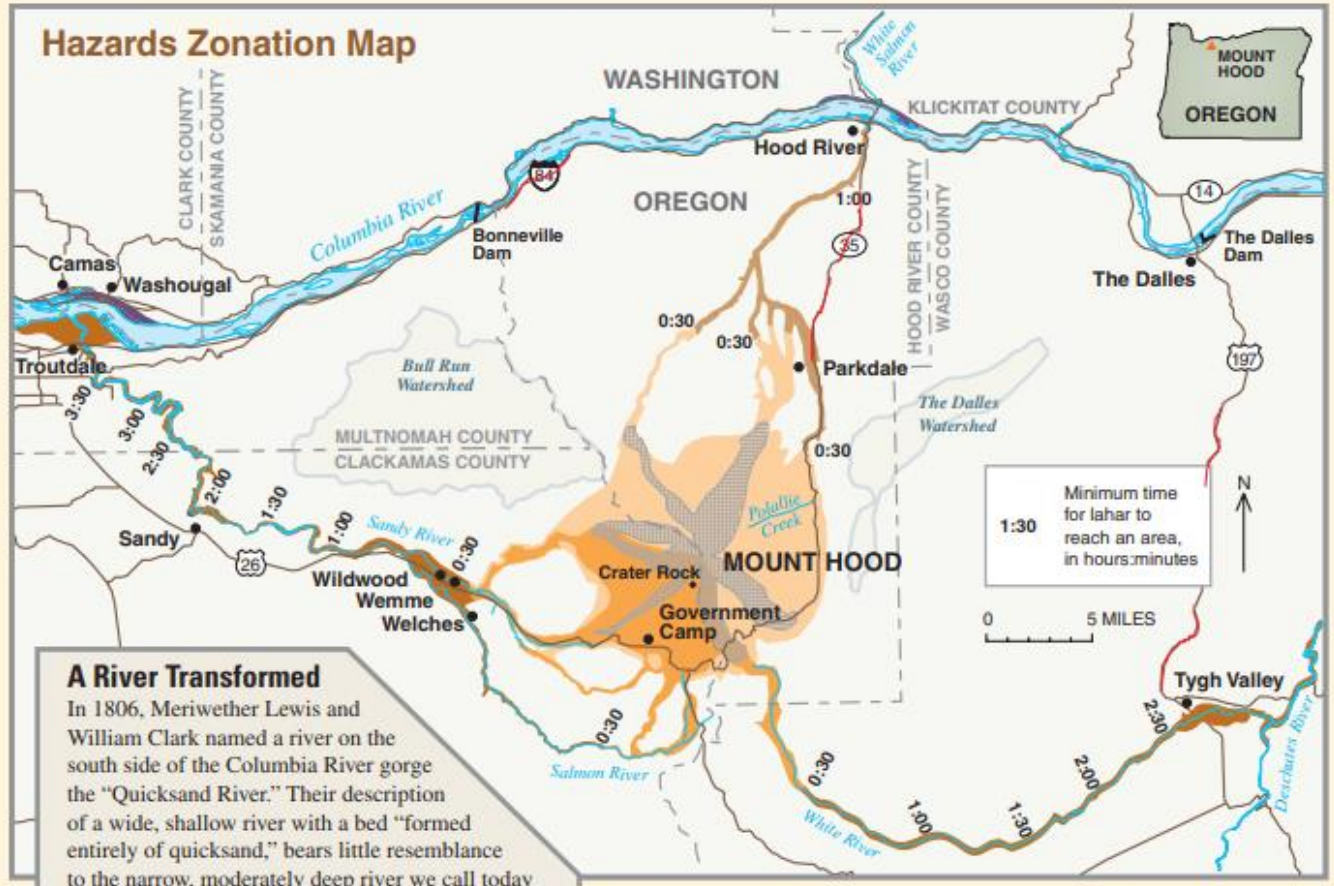
Explosive eruptions blast lava fragments (**tephra**) and gas into the air. Tephra can also be carried aloft in billowing ash clouds above pyroclastic flows. Large fragments fall to the ground close to the volcano, but smaller fragments (**ash**) can travel hundreds to thousands of miles downwind.

Debris Avalanches and Lahars

Debris avalanches are rapid landslides of rock, soil and overlying vegetation, snow, or ice. They can bury or smash objects in their path. All or some portion of debris avalanches can transform into lahars. **Lahars** are fast-moving slurries of rock, mud, and water that move down river valleys. They can bury, move, or smash objects in their path. Lahars form when pyroclastic flows melt snow or ice, or by the transformation of debris avalanches, or by mobilization of loose debris on the flanks of volcanoes.



Hazards Zonation Map



A River Transformed

In 1806, Meriwether Lewis and William Clark named a river on the south side of the Columbia River gorge the "Quicksand River." Their description of a wide, shallow river with a bed "formed entirely of quicksand," bears little resemblance to the narrow, moderately deep river we call today the Sandy River. What happened? The answer lay 50 miles away at Mount Hood. An eruption in the 1790's caused a tremendous amount of volcanic rock and sand to enter the Sandy River drainage. That sediment was still being flushed downstream when Lewis and Clark saw and named the river. Since 1806, the river has removed the excess sediment from its channel. The Toutle River in southwest Washington was similarly affected by the 1980 eruptions of Mount St. Helens. A sediment-retention dam was built in the late 1980's to trap sediment and ease flooding in the lower Toutle River below the dam.

Proximal Hazard Zones (P)
(lava flow, pyroclastic flow, debris avalanche, lahar)

PA PB

Distal Hazard Zones (D)
(lahar)

DA DB

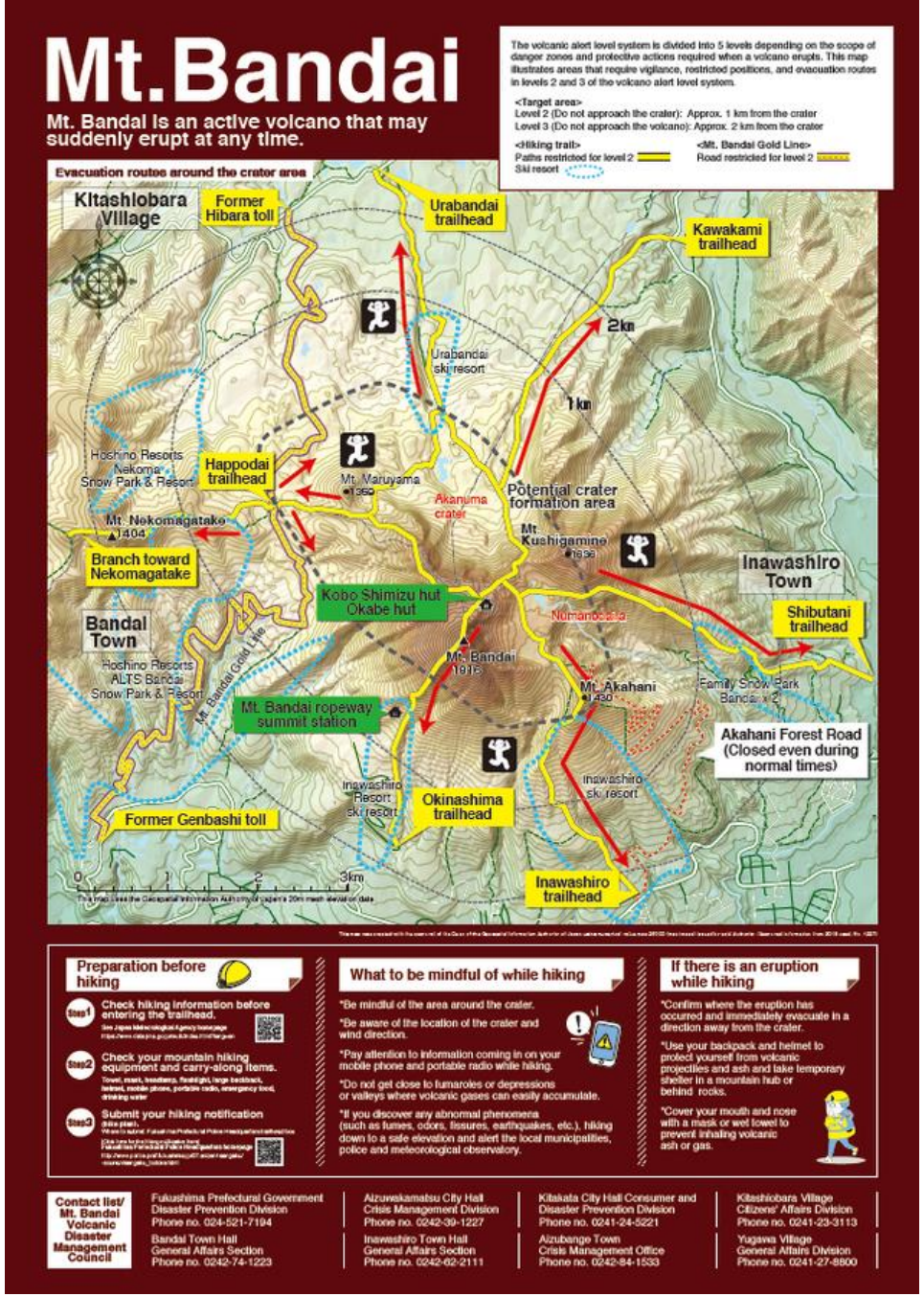
Bank Erosion

Major valleys

Simplified volcano-hazards-zonation map for Mount Hood, Oregon. For more information on hazard zones see text or for a more detailed map and description see Scott and others, 1997, Volcano Hazards in the Mount Hood Region, Oregon, U.S. Geological Survey Open-File Report 97-89.

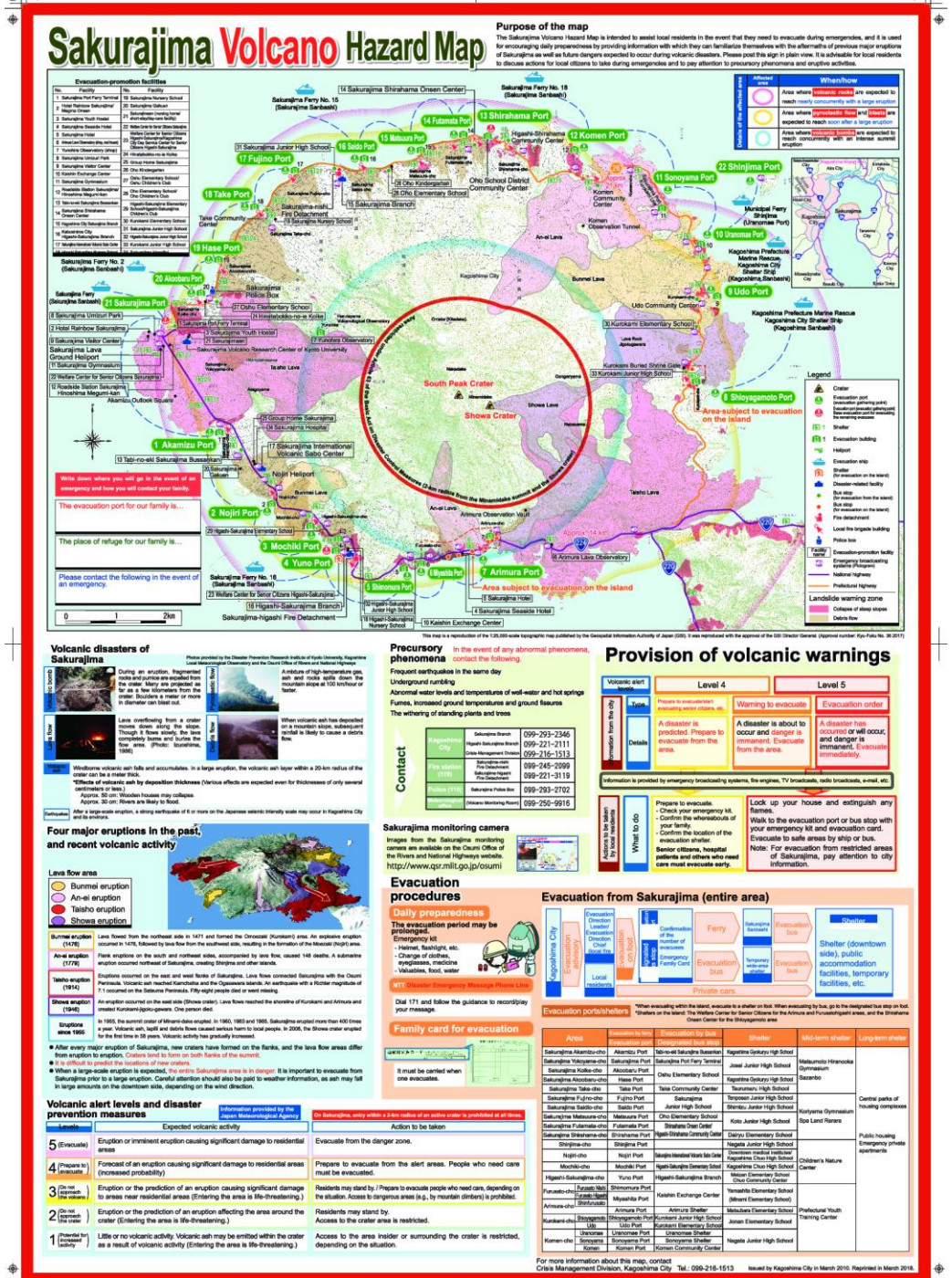


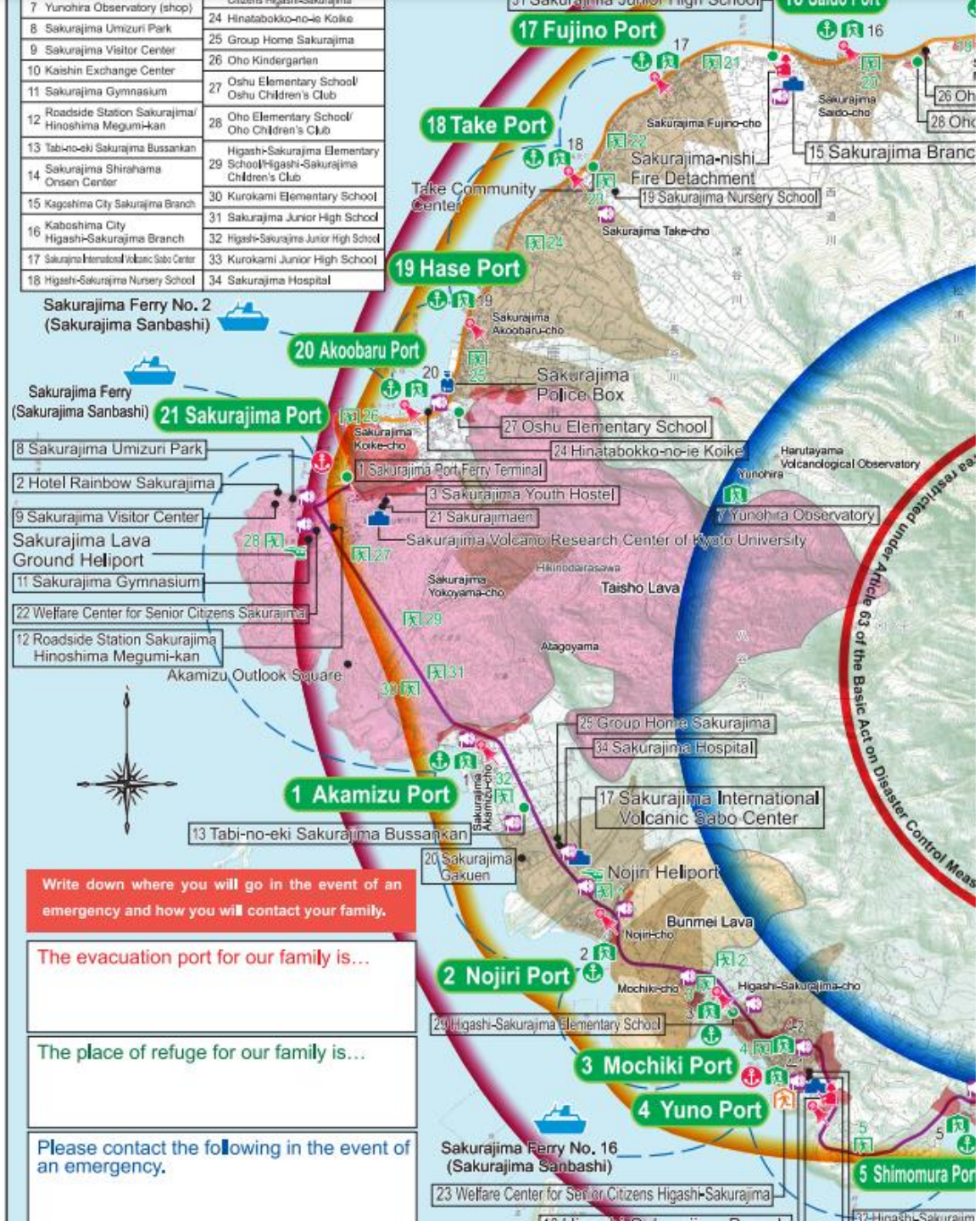
Ruapehu, New Zealand (GNS Science 2021)



Bandai, Japan (Bandaisan Volcanic Disaster Management Council, 2019)

Sakurajima (Aira), Japan (Kagoshima City 2018)





7	Yunohira Observatory (shop)
8	Sakurajima Umizuri Park
9	Sakurajima Visitor Center
10	Kaishin Exchange Center
11	Sakurajima Gymnasium
12	Roadside Station Sakurajima/ Hinoshima Megumi-kan
13	Tabi-no-eki Sakurajima Bussankan
14	Sakurajima Shirahama Onsen Center
15	Kagoshima City Sakurajima Branch
16	Kaboshima City Higashi-Sakurajima Branch
17	Sakurajima International Volcanic Sabo Center
18	Higashi-Sakurajima Nursery School
24	Hinatabokko-no-ie Koike
25	Group Home Sakurajima
26	Oho Kindergarten
27	Oshu Elementary School/ Oshu Children's Club
28	Oho Elementary School/ Oho Children's Club
29	Higashi-Sakurajima Elementary School/Higashi-Sakurajima Children's Club
30	Kurokami Elementary School
31	Sakurajima Junior High School
32	Higashi-Sakurajima Junior High School
33	Kurokami Junior High School
34	Sakurajima Hospital

Evacuation procedures

Daily preparedness

The evacuation period may be prolonged.

Emergency kit

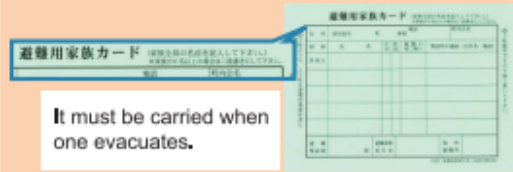
- Helmet, flashlight, etc.
- Change of clothes, eyeglasses, medicine
- Valuables, food, water



NTT Disaster Emergency Message Phone Line

Dial 171 and follow the guidance to record/play your message.

Family card for evacuation



It must be carried when one evacuates.

Evacuation ports/shelters

*When evacuating within the island, evacuate to a shelter on foot. When ev
*Shelters on the island: The Welfare Center for Senior Citizens for the Ari
Onsen Center for the Shioyagamoto area

Area	Evacuation by ferry: Evacuation port	Evacuation by bus: Designated bus stop	Shelter
Sakurajima Akamizu-cho	Akamizu Port	Tabi-no-eki Sakurajima Bussankan	Kagoshima Gyokuryu High School
Sakurajima Yokoyama-cho	Sakurajima Port	Sakurajima Port Ferry Terminal	Josei Junior High School
Sakurajima Koike-cho	Akoobaru Port	Oshu Elementary School	
Sakurajima Akoobaru-cho	Hase Port		Kagoshima Gyokuryu High School
Sakurajima Take-cho	Take Port	Take Community Center	Tsurumaru High School
Sakurajima Fujino-cho	Fujino Port	Sakurajima Junior High School	Tenposan Junior High School
Sakurajima Saidio-cho	Saido Port		Shimizu Junior High School
Sakurajima Matsuura-cho	Matsuura Port	Oho Elementary School	Koto Junior High School
Sakurajima Futamata-cho	Futamata Port	Shirahama Onsen Center/ Higashi-Shirahama Community Center	
Sakurajima Shirahama-cho	Shirahama Port		Dairyu Elementary School
Shinjima-cho	Shinjima Port		Nagata Junior High School
Nojiri-cho	Nojiri Port	Sakurajima International Volcanic Sabo Center	Downtown medical institutes/ Kagoshima Chuo High School
Mochiki-cho	Mochiki Port	Higashi-Sakurajima Elementary School	Kagoshima Chuo High School
Higashi-Sakurajima-cho	Yuno Port	Higashi-Sakurajima Branch	Meizan Elementary School Chuo Community Center
Furusato-cho: Furusato Nishi	Shimomura Port	Kaishin Exchange Center	Yamashita Elementary School (Minami Elementary School)
Furusato-cho: Furusato Higashi / Arimura-cho: Shinfurusato	Miyashita Port		
Arimura-cho	Arimura Port	Arimura Shelter	Matsubara Elementary School
	Shioyagamoto Port	Kurokami Junior High School	

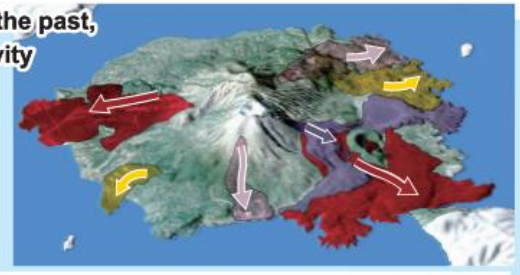
Volcanic alert levels and disaster prevention measures

Information provided by the Japan Meteorological Agency

On Sakurajima, entry within a 2-km radius of an active crater is prohibited at all times.

Levels	Expected volcanic activity	Action to be taken
5 (Evacuate)	Eruption or imminent eruption causing significant damage to residential areas	Evacuate from the danger zone.
4 (Prepare to evacuate)	Forecast of an eruption causing significant damage to residential areas (increased probability)	Prepare to evacuate from the alert areas. People who need care must be evacuated.
3 (Do not approach the volcano)	Eruption or the prediction of an eruption causing significant damage to areas near residential areas (Entering the area is life-threatening.)	Residents may stand by. / Prepare to evacuate people who need care, depending on the situation. Access to dangerous areas (e.g., by mountain climbers) is prohibited.
2 (Do not approach the crater)	Eruption or the prediction of an eruption affecting the area around the crater (Entering the area is life-threatening.)	Residents may stand by. Access to the crater area is restricted.
1 (Potential for increased activity)	Little or no volcanic activity. Volcanic ash may be emitted within the crater as a result of volcanic activity (Entering the area is life-threatening.)	Access to the area insider or surrounding the crater is restricted, depending on the situation.

Four major eruptions in the past, and recent volcanic activity



Lava flow area

- Bunmei eruption
- An-ei eruption
- Taisho eruption
- Showa eruption

Bunmei eruption (1476)	Lava flowed from the northeast side in 1471 and formed the Omoezaki (Kurokami) area. An explosive eruption occurred in 1476, followed by lava flow from the southwest side, resulting in the formation of the Moezaki (Nojiri) area.
An-ei eruption (1779)	Flank eruptions on the south and northeast sides, accompanied by lava flow, caused 148 deaths. A submarine eruption occurred northeast of Sakurajima, creating Shinjima and other islands.
Taisho eruption (1914)	Eruptions occurred on the east and west flanks of Sakurajima. Lava flows connected Sakurajima with the Osumi Peninsula. Volcanic ash reached Kamchatka and the Ogasawara islands. An earthquake with a Richter magnitude of 7.1 occurred on the Satsuma Peninsula. Fifty-eight people died or went missing.
Showa eruption (1946)	An eruption occurred on the east side (Showa crater). Lava flows reached the shoreline of Kurokami and Arimura and created Kurokami-jigoku-gawara. One person died.
Eruptions since 1955	In 1955, the summit crater of Minami-dake erupted. In 1960, 1983 and 1985, Sakurajima erupted more than 400 times a year. Volcanic ash, lapilli and debris flows caused serious harm to local people. In 2006, the Showa crater erupted for the first time in 58 years. Volcanic activity has gradually increased.

- After every major eruption of Sakurajima, new craters have formed on the flanks, and the lava flow areas differ from eruption to eruption. Craters tend to form on both flanks of the summit.
- It is difficult to predict the locations of new craters.
- When a large-scale eruption is expected, the entire Sakurajima area is in danger. It is important to evacuate from Sakurajima prior to a large eruption. Careful attention should also be paid to weather information, as ash may fall in large amounts on the downtown side, depending on the wind direction.

Volcanic disasters of Sakurajima

Photos provided by the Disaster Prevention Research Institute of Kyoto University, Kagoshima Local Meteorological Observatory and the Osumi Office of Rivers and National Highways

Volcanic bomb



During an eruption, fragmented rocks and pumice are expelled from the crater. Many are projected as far as a few kilometers from the crater. Boulders a meter or more in diameter can blast out.

Pyroclastic flow



A mixture of high-temperature gas, ash and rocks spills down the mountain slope at 100 km/hour or faster.

Lava flow



Lava overflowing from a crater moves down along the slope. Though it flows slowly, the lava completely burns and buries the flow area. (Photo: Izuoshima, 1986)

Debris flow



When volcanic ash has deposited on a mountain slope, subsequent rainfall is likely to cause a debris flow.

Volcanic ash

Windborne volcanic ash falls and accumulates. In a large eruption, the volcanic ash layer within a 20-km radius of the crater can be a meter thick.

***Effects of volcanic ash by deposition thickness** (Various effects are expected even for thicknesses of only several centimeters or less.)

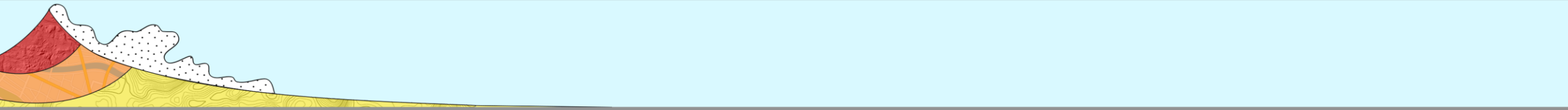
Approx. 50 cm: Wooden houses may collapse.
Approx. 30 cm: Rivers are likely to flood.

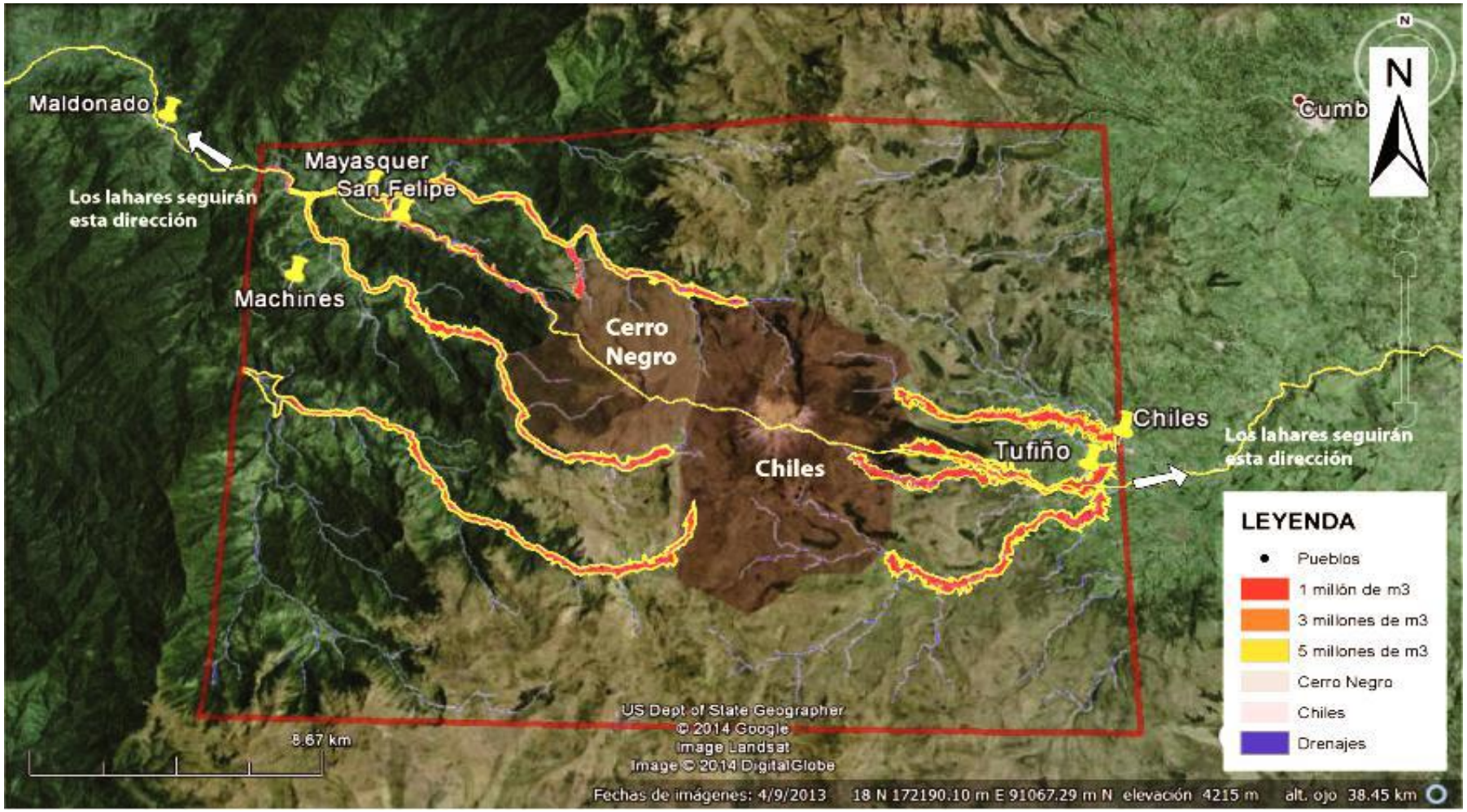
Earthquakes

After a large-scale eruption, a strong earthquake of 6 or more on the Japanese seismic intensity scale may occur in Kagoshima City and its environs.

Models are just one tool!

- Models are just one tool to help you make a hazard map
- The model outputs can be modified using your geological knowledge about the volcano and about the hazards, what is known about past events, and any other additional knowledge

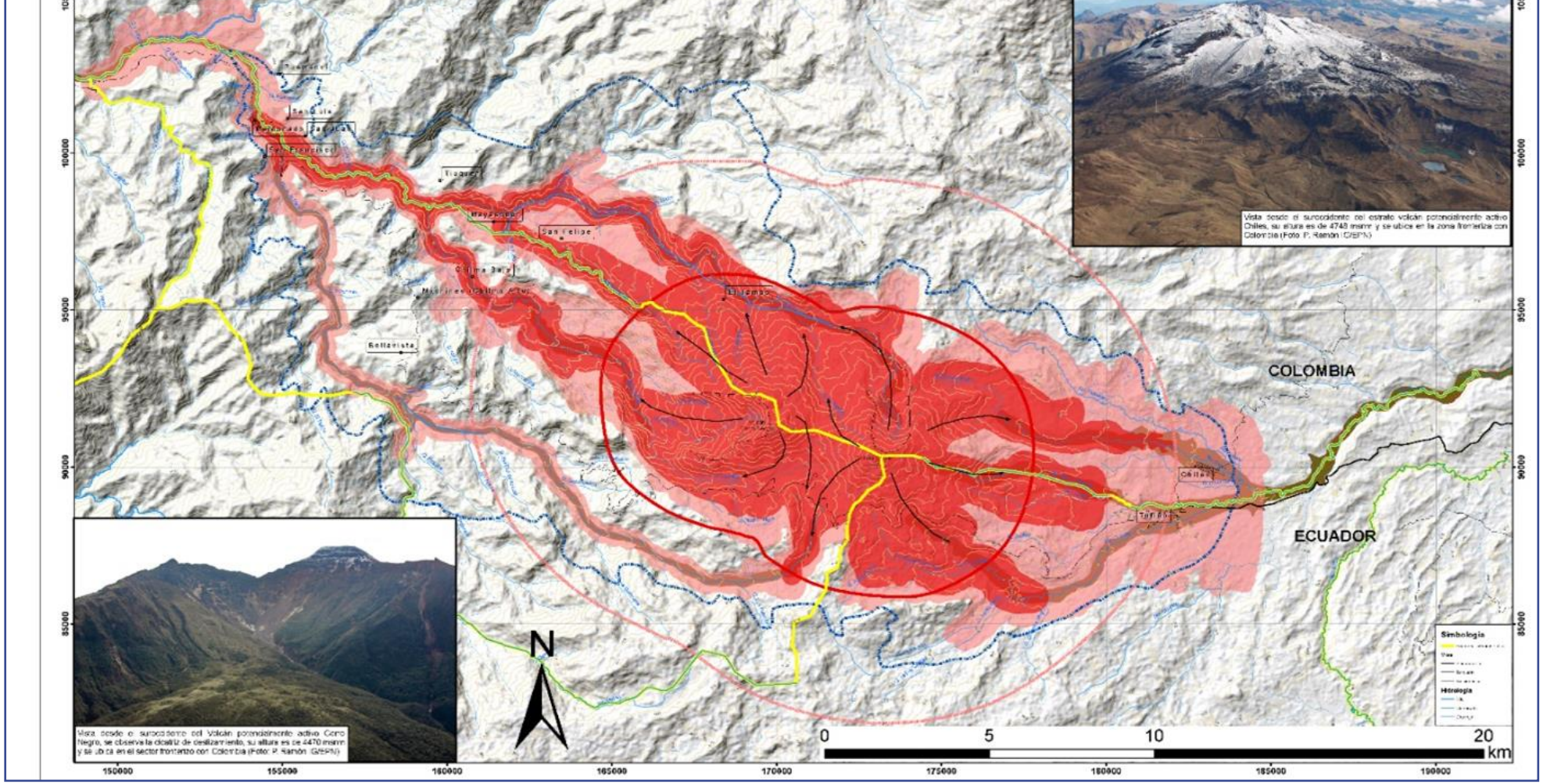




Chiles-Cerro Negro, Colombia-Ecuador border

Software LaharZ. Casos de erupción del Cerro Negro o Chiles.
P. Espin y F. Vasconez. IGEPN

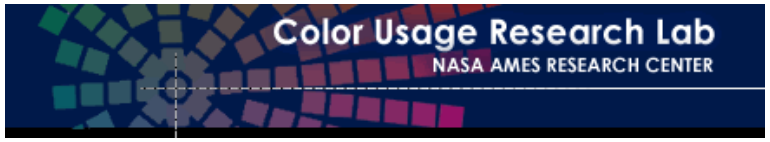
USGS VDAP USAID FROM THE AMERICAN PEOPLE



Chiles-Cerro Negro, Colombia-Ecuador border

Web resources

betterfigures.org
climate-lab-book.ac.uk
http://colorbrewer2.org/
colorusage.arc.nasa.gov NASA AMES Color Usage Research Lab
https://eagereyes.org/basics/rainbow-color-map



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